CHENNAI - MONDAY - MAY 4, 2026

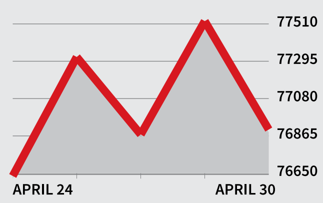
₹10 • Pages 10 • Volume 33 • Number 123

www.businessline.in

businessline thehindubusinessline

the hindu businessline.

SENSEX 76913.50 (+249.29)



IN FOCUS

	Week's close	Week's change
Nifty 50	23997.55	+99.60
P/E Ratio (Sensex)	20.94	-0.04
US Dollar (in ₹)	94.91	+0.67
Gold Std 10 gm (in ₹)	149661.00	-1211
Silver 1 kg (in ₹)	240331.00	-3494

TAKING STOCK.
Nestle India CMD Manish Tiwary discusses cost pressures, category growth drivers and rural salience **p3**

SNAG PERSISTS.
Govt and electronics industry are still under discussion on Six Sigma and local sourcing **p10**

BENGALURU - CHENNAI - COIMBATORE - HUBBALLI - HYDERABAD - KOCHI - KOLKATA - MADURAI - MALAPPURAM - MANGALURU - MUMBAI - NOIDA - THIRUVANANTHAPURAM - TIRUCHIRAPALLI - VIJAYAWADA - VISAKHAPATNAM

Regd. TN/ARD/14/2012-2014, RNI No. 55320/94

MONDAY SPECIALS.

QUANTUM

Indian material for magnet making

Can India achieve magnet independence? The honest answer is, no — at least, not anytime soon. However, it can reduce its dependence on other countries, especially China, for its magnet requirement. Now, Raghavan Gopalan, veritably the "magnet man of India", suggests that India can reduce its import dependence for magnets by using one rare earth mineral abundantly available within its shores — cerium. **p7**

LOGISTICS

Unruly shrub fuels Kandla's eco goals

In the dry stretches around Deendayal Port Authority (formerly Kandla port) in Gujarat, *Prosopis juliflora*, locally known as *gando bawal*, has long been viewed as a nuisance. The invasive shrub spreads rapidly, degrades land and crowds out native vegetation. Now, it is being recast as a resource at the heart of a bio-methanol project at the port to convert waste biomass into clean fuel. **p8**

Banks may sell written-off loans to fund new ECL norms

ASSET RESOLUTION. Sale to ARCs seen unlocking capital for expected credit loss shift

K Ram Kumar
Mumbai

POTENTIAL GOLDMINE

- As of March-end 2025, scheduled commercial banks' pool of GNPAs stood at ₹4,19,099 crore
- The pool of TWO accounts is estimated to be at least double the GNPA amount
- TWO accounts refer to loans that have remained in the non-performing category for four years or more and for which full provisioning has been made

Sale of a portion of the huge pile of technical write-off accounts to asset reconstruction companies (ARCs) could emerge as a linchpin for banks to unlock funds to make provisions for loans under the ECL (expected credit loss)-based regime, which will come into effect from April 1, 2027.

As at March-end 2025, while scheduled commercial banks' pool of gross non-performing assets (GNPAs) stood at ₹4,19,099 crore, the pool of technical write-off (TWO) accounts is estimated to be at least double the GNPA amount. Up to 40 per cent of the TWO accounts have the potential to be recovered, according to estimates.

RECOVERY PROCESS

Referring to the Finance Ministry's nudge to banks three years ago to step up recovery from written-off accounts from just 14 per cent then to 40 per cent, a senior public sector bank official said the time is now ripe for fast-tracking recovery from TWO accounts in the backdrop of the requirement for making provisioning as per the ECL norms.

TWO accounts refer to loans that have remained in the non-performing category for four years or more and for which full provisioning has been made. So, such accounts are a potential goldmine that banks could harness to make provisions under the ECL framework.

"There is a huge pool of technical write-off accounts off the balance sheet of banks, possibly larger than NPAs appearing on their balance sheet. Recovery in these accounts goes straight to banks' bottom line. So, sale of such accounts to ARCs can be explored as an option to meet additional provisioning requirements on account of migration to the ECL framework," said Hari Hara Mishra, CEO, Association of ARCs in India.

Currently, banks make provisioning only when a default occurs under the incurred-loss-based provisioning framework. However, next year, banks will move to the ECL-based provisioning norm, whereby they have to estimate and provide for potential future credit losses before they actually occur.

FOCUS ON TWO A/Cs

Alluding to NPAs hitting multi-decadal lows, with GNPAs at 2.2 per cent and net NPAs at 0.5 per cent at September-end 2025, the chief of an asset reconstruction company said unlike earlier phases when ARCs focused on newly classified NPAs, the focus had now shifted decisively toward fully written-off accounts. "The rationale for selling fully provisioned written-off assets is straightforward. Bank managements prefer to focus their bandwidth on the performing 98 per cent of the book rather than the distressed 2 per cent, which are hard nuts to crack. Recoveries from sale of such assets result in a direct write-back to profits," the ARC chief said.

Crisil Ratings, in a recent report, assessed ECL norms to have a one-time net impact of up to 120 basis points on banks' Common Equity Tier-1 ratio. However, they get to defray this cost across four fiscals, while additional advance provisioning can also reduce the impact.

In its circular on compromise settlements and technical write-offs, the RBI said TWO is a normal banking practice undertaken by lenders to cleanse balance sheets of bad debts, either considered unrecoverable or whose recovery is likely to consume disproportionate resources of the lenders.

Mittal family-Adar Poonawalla to buy Rajasthan Royals for $1.65 b

Meenakshi Verma Ambwani
New Delhi

ROYAL TAKEOVER

- Post transaction, the Mittal family will own 75% stake, while Adar Poonawalla will hold an 18% stake
- Transaction expected to close in Q3 CY2026 after regulatory approvals
- The earlier proposed buyout of the franchise by the Kal Somani-led consortium fell through due to funding constraints and regulatory complexities

Steel magnate Lakshmi N Mittal's family, in partnership with Adar Poonawalla, is set to acquire a controlling stake in the IPL team Rajasthan Royals.

The transaction consideration is about $1.65 billion and represents the enterprise value of Rajasthan Royals' men's franchise and associated teams, Paarl Royals in South Africa and Barbados Royals in the Caribbean.

"Lakshmi N Mittal and Aditya Mittal today announce that a definitive agreement has been reached to acquire the Rajasthan Royals, in partnership with Adar Poonawalla, from Manoj Badale and consortium," an official statement said.

AWAITS APPROVALS

The transaction is expected to close in the third quarter of 2026, subject to regulatory approvals including from BCCI, the CCI and the IPL Governing Council.

As per the deal, the Mittal family will hold about 75 per cent in the IPL franchise, while Poonawalla will own about 18 per cent. Existing "approved" investors, including Badale, will hold the remaining 7 per cent stake.

The restructured board will include Lakshmi N Mittal, Aditya Mittal, Vanisha Mittal-Bhatia, Poonawalla and Badale.

The announcement comes after the earlier proposed buyout of the franchise by the Kal Somani-led consortium, which included global investors such as Rob Walton and Sheila Ford Hamp, fell through due to funding constraints and regulatory complexities, besides some disagreements.

"I love cricket and my family is from Rajasthan, so there is no IPL team that I would rather be part of than the Rajasthan Royals... I look forward to being part of this great team, and cannot wait to join all the fans at the pitch-side to cheer on our future success," Lakshmi N Mittal, Executive Chairman, ArecelorMittal, said in a statement.

'STRONG LEGACY'

"Rajasthan Royals is a premier IPL franchise with a strong legacy, and I look forward to supporting its continued growth and long-term success," Poonawalla, CEO, Serum Institute of India said.

Both the Mittal family and Poonawalla had reportedly made bids to buy Rajasthan Royals and Royal Challengers Bengaluru.

"We are delighted to welcome the Mittal family and Adar Poonawalla as the new owners of the Rajasthan Royals. Their passion for cricket, their connection to Rajasthan and India, and their long-term ambition for the franchise make them ideal custodians of the next chapter," Badale added.

Foreign airlines trim India flights as fuel costs rise

Aneesh Phadnis
Mumbai

ROUTE RESET. While Australia's Qantas is temporarily suspending flights between Sydney and Bengaluru, Thai AirAsia is cutting flights to multiple Indian cities

International airlines, including those from Australia and Thailand, are reducing their flights to India amid soaring jet fuel prices.

While Australia's Qantas is temporarily suspending flights between Sydney and Bengaluru between August and October-end, Thai AirAsia is cutting flights to multiple Indian cities.

Qantas said it continues to take action to mitigate the impact of the conflict in West Asia that has increased fuel costs.

While it has redeployed capacity to Europe to cater to the travel demand, the airline has reduced flights to India. Qantas has also reduced frequency on domestic and New Zealand routes.

AirAsia Thailand is suspending flights from Phuket to Chennai and Kochi, besides cancelling flights to other Indian cities.

Overall, Thai AirAsia has cut capacity by 30 per cent. On the international side, the cuts primarily impact India, where elevated prices have reduced its ability to offer low fares, the airline said.

Domestic and international carriers have introduced or hiked fuel surcharges to cover up the increase in fuel prices.

While the price hike for scheduled domestic flights is limited to 25 per cent, jet fuel for international flights now costs double of the March level.

As such travel demand too is uncertain and popular South-East Asian routes haven't seen a pickup. This has resulted in Air India Express reducing flights to Thailand in May.

"Demand for international leisure travel has slowed down, and customers are thinking twice before booking trips," said Anil Kalsi, Vice-President of the Travel Agents Federation of India.

AI CUTS FREQUENCY

Air India is reducing frequency on some Europe and North America sectors by 10-12 per cent to mitigate the challenge.

On the East Asia side, reduction is planned on the upcoming Mumbai-Tokyo route in July.

In a staff email on Friday, Air India CEO and Managing Director Campbell Wilson said the airline had cut flights in April and May as rising fuel costs made the operations unprofitable.

"The airspace and jet-fuel price situation remains extremely challenging, leaving us with no choice but to further trim schedules for June and July," Wilson said.

CALM BEFORE THE STORM

BALLOT BUZZ. With high expectations surrounding the electoral debut of C Joseph Vijay, actor and leader of the Tamilaga Vettri Kazhagam, people were seen thronging his residence in Chennai on Sunday for selfies. Votes for the Assembly elections in Tamil Nadu, Assam, Kerala, Puducherry and West Bengal will be counted on Monday, with the results set to shape the fortunes of parties such as TMC, DMK, AIADMK, BJP, Congress and the Left **(Report on page 2)** BIJOY GHOSH

Bengaluru feels the heat of climate change

Rising temperatures and warmer nights redefine the Garden City's weather

Adithya Lalgudi
Bengaluru

WORRYING FACTORS. Between 1973 and 2023, the city's built-up area expanded by 1,055%, leading to an 88% loss of vegetation and a 79% fall in water bodies, according to a study

Once a retirement paradise known for its mild climate, Bengaluru is now feeling the heat. With summer intensifying, the city recorded a maximum temperature of 36°C in April, according to the India Meteorological Department, while the average minimum temperature rose to 23.55°C.

The rise in night-time temperatures, experts say, is particularly concerning.

"The worrying factor is that the minimum temperature has gone up, and the duration of heat has extended," said Viswanath Srikantaiah, Founder of Rainwater Club. He attributed the trend partly to the urban heat island effect.

Experts said the shift is becoming more visible on the ground. "The heat arrived much earlier and was sharper this year; you could feel it, even in the surge of air conditioner use across the city. While annual variations will continue, the overall trajectory in a business-as-usual scenario is clearly upwards in terms of rising temperatures," said Jaya, Director, WRI India.

A 2025 study, titled *Urban Heat Island in Bengaluru: Built-up Growth and Temperature Trends*, highlights the scale of the shift. Between 1973 and 2023, the city's built-up area expanded by 1,055 per cent, leading to an 88 per cent loss of vegetation and a 79 per cent decline in water bodies. This contributed to a 7.9°C rise in land surface temperatures, establishing a clear link between rapid concretisation and increasing heat.

"We are compounding climate change by over-concretising the city," said urbanist V Ravichandar.

WATER OUTLOOK

Despite lingering concerns over water shortages, fuelled by last year's crisis, there is no immediate cause for alarm. According to Srikantaiah, the city has sufficient water supply at least until the first week of June, supported by expanding distribution networks.

Groundwater depletion tends to be seasonal and localised. "In some pockets it remains a concern, but overall, it is not a city-wide issue," he said. While the IMD expects rainfall to be below the long-period average, he said even modest monsoon showers should be sufficient to replenish the reservoirs.

(The writer is an intern)

Space start-up GalaxEye launches the 'world's first' all-weather imaging satellite

Our Bureau
Bengaluru

Mission Drishti is India's largest privately developed Earth observation satellite

Bengaluru-based GalaxEye has launched the world's first OptoSAR satellite, marking a significant milestone for India's emerging private space ecosystem. The satellite, Mission Drishti, was launched aboard a SpaceX Falcon 9 rocket from Vandenberg, California.

Weighing 190 kg, it is India's largest privately developed Earth observation satellite, and the first globally to integrate synthetic aperture radar (SAR) and electro-optical (EO) sensors on a single platform.

This combination enables all-weather, day-and-night imaging, overcoming long-standing limitations of conventional systems that struggle with cloud cover or low-light conditions. The dual-use satellite can support applications across agriculture, defence, disaster management, maritime monitoring, and infrastructure planning.

PAT FROM PM

Prime Minister Narendra Modi congratulated the start-up. "Mission Drishti by GalaxEye marks a major achievement in our space journey. The successful launch of the world's first OptoSAR satellite, and the largest privately-built satellite in India is a testament to our youth's passion for innovation and nation-building. Heartiest congratulations and best wishes to the founders and the entire team of GalaxEye," he said on X.

The launch comes at a time when India is strengthening its position in the global geospatial intelligence market. With 29 active Earth observation satellites, according to ISRO, the country is increasingly seen as a reliable provider of intelligence, surveillance and reconnaissance (ISR) capabilities for both domestic and international users.

Industry stakeholders say the mission underscores the growing maturity of India's private space sector.

TANGIBLE OUTCOMES

Pavan Goenka, Chairman of IN-SPACe, said sustained efforts in commercialisation and capacity building over the past five to six years were now translating into tangible outcomes.

GalaxEye plans to scale up its satellite constellation over the next five years to strengthen Earth observation infrastructure and meet rising global demand for satellite data. It has partnered with NewSpace India Ltd to distribute its imagery globally, a move that is expected to expand access to high-resolution geospatial data across industries.

CHENNAI
businessline.
MONDAY - MAY 4 - 2026

QUICKLY.

Ramagundam coal mine project gets green nod

Hyderabad: The Ramagundam coal mine project of Singareni Collieries Company (SCCL) has received a nod from the Centre. The Hyderabad-based firm has got information about primary environmental, forest and climatic clearances and the formal documents will be received shortly, according to a release. The project aims to produce 210 lakh tonnes of coal annually. The coal produced from the project will be supplied to NTPC and other coal-based industries in the region, the release said. OUR BUREAU

Telangana to release ₹6,000 cr retirement dues

Hyderabad: In a move that could bring relief to retired State government employees in Telangana, Chief Minister A Revanth Reddy has directed officials to clear pending retirement benefits worth ₹6,000 crore over the next 100 days. Reddy, who held talks with the representatives of employees, also asked the pay revision panel for government staff to submit its report at the earliest. OUR BUREAU

India may take middle path, unlikely to remove import duty on cotton

ON THE TABLE. Either zero duty import in Sept-Oct period or reduction to 6-7% from the current 11%

Prabhudatta Mishra
Amiti Sen
New Delhi

BALANCING OUTPUT. Officials in Punjab said cotton sowing is nearly over. In Gujarat, Madhya Pradesh and Maharashtra, planting starts in May and in the south from July NAGARA GOPAL

Union Agriculture Minister Shivraj Singh Chouhan suggested a middle path amid a demand from the Ministry of Textiles and the industry to allow cotton import at zero duty till December 2026, which was opposed by the Ministry of Agriculture and Farmers' Welfare on the ground that it would send a wrong signal to farmers as sowing has already begun.

Options under a middle path could be either zero duty import in September-October as current stock in the country is comfortable to meet domestic demand at least till August. The other option could be to reduce import duty to 6-7 per cent from the current 11, official sources said.

Chouhan, last week, held a meeting with representatives of the Federation of Indian Export Organisations (FIEO) and officials from the Textiles, Agriculture, Commerce and Finance Ministries to examine the demand of garment exporters.

Sources said the Minister was informed that an estimated 47 lakh bales of cotton, as of now, were available with the public sector Cotton Corporation of India, and that together with the stock held by the private sector, including processors and traders, the domestic demand of the industry till August could be easily met.

LOCAL PRICES

Since cotton is already being imported, the only issue is its domestic price, which the industry says is quite high now.

Sources said that as new harvests will start arriving from October, sufficient stock should be available in the country for the September-October period.

Officials in Punjab said that cotton sowing is nearly over, after it started from around mid-April. In major growing States like Gujarat, Madhya Pradesh and Maharashtra, planting starts in May and in the southern States from July.

COTTON ACREAGE DIPS

In 2025-26, cotton production dipped to 290.91 lakh bales (of 170 kg/bale) from 297.24 lakh bales in 2024-25 despite the acreage remaining stagnant at 114.8 lakh hectares in the last two years, after dropping from a high of 129.27 lakh hectares in 2022-23.

More assured returns from maize and paddy, as well as crop risk from the pink bollworm (PBW) pest, are driving cotton farmers to reduce acreage. In the past six years, cotton area has shrunk the maximum — by 20 lakh hectares – among all other crops, data show.

According to the Cotton Association of India (CAI) estimates, in the 2025-26 season (October-September), the country's production may be 324 lakh bales, consumption 315 lakh bales, import 47 lakh bales and export 15 lakh bales. The trade body pegged the closing stock on March 31 at nearly 292 lakh bales.

As reported by *businessline* earlier, the Textiles Ministry had pitched for a temporary removal or reduction in import duties on raw cotton to reduce input costs for apparel exporters amid the West Asia crisis. Rising yarn and fabric prices have spiked production costs, eroding the industry's export competitiveness despite strong global demand, the AEPC had said.

While India is one of the world's largest producers of cotton, it also imports to meet demand during lean summer months and also source superior varieties not available locally. Textiles and garment exports fell 2.2 per cent to $35.8 billion in 2025-26 due to a contraction in shipments of key segments such as cotton, per an analysis of the latest trade data by research body GTRI.

Will the power pendulum swing? Result day for WB, Assam, TN, Kerala, Pondy

KEEPING VIGIL. Security personnel outside a strong room in Kolkata on Sunday, on the eve of counting for the West Bengal Assembly polls PTI

Press Trust of India
New Delhi/Kolkata/Chennai

Counting of votes will be held on Monday in Tamil Nadu, Assam, Kerala, Puducherry and West Bengal after a riveting high-voltage contest with the outcome crucial for prominent ruling regional parties like the TMC and the DMK as well as the BJP, the Congress and the Left.

The exercise will begin at 8 am, starting with postal ballots, at the counting centres, which will have a three-tier security system. In a first, the Election Commission has introduced a QR code-based photo identity card system through ECINET to prevent unauthorised entry into the counting centres.

ARRANGEMENTS MADE

Votes will be counted across 77 centres for 293 Assembly seats in West Bengal, which witnessed unprecedented security deployment this time and an acrimony-filled run-up to the result day, with both the ruling TMC and the opposition BJP expressing apprehensions of vote manipulation.

The two-phase polls in the State ended on April 29, with its highest-ever voter turnout of 92.47 per cent since Independence.

The election was countermanded in one constituency, Falta, in the South 24 Parganas district due to "severe electoral offences", and fresh polls will be held there on May 21.

The BJP-led NDA is hoping for a hat-trick in Assam.

Twenty-five companies (around 100 personnel each) of the Central Armed Police Forces (CAPF) have been deployed to guard the counting centres and the strongrooms housing the electronic voting machines (EVMs), officials said.

Polling in the State was held on April 9, with 85.96 per cent turnout.

In Kerala, buoyed by its performance in the 2024 Lok Sabha polls and recent local body elections, the Congress-led UDF is hoping to oust the Left Democratic Front (LDF), which has been in power for two terms, while the BJP-led NDA is trying to gain a foothold in the State with a bipolar policy.

The ruling DMK in Tamil Nadu is hoping for a second consecutive term after contesting in a changed political arena as, apart from its main rival AIADMK, there were new entrants like actor-politician Vijay-led TVK's and also Tamil nationalist Seeman's NTK.

Approximately 1.25 lakh personnel, who include officials and micro-observers on vote-counting duty and police, have been deployed.

In Puducherry, six counting centres have been set up across the Union Territory.

Brokerages close FY26 on strong Q4 rebound led by retail churn, volatility

Akshata Gorde
Mumbai

India's leading brokerage houses capped FY26 with a strong March quarter performance, riding a rebound in retail trading activity, improved market volatility and operating leverage from digital-heavy models, even as the full-year picture remained uneven.

BAGGING THE BUCKS

Among the key players, Groww, Angel One, HDFC Securities and Kotak Securities reported healthy year-on-year growth in the March quarter, a strong recovery from a relatively softer first half.

Groww topped the chart, reporting a sharp jump in both revenue and profit for Q4, aided by strong client additions and improving operating leverage.

The company said, "Operating leverage played out across all the cost buckets... margins will keep expanding as revenue increases faster than costs."

The user base crossed 2.1 crore, with client assets rising 35 per cent on-year, underlining continued retail participation despite intermittent market volatility.

Angel One also posted strong earnings growth, with net profit at ₹320.2 crore and revenue at ₹1,467.2 crore for the quarter, driven by higher trading volumes, retail activity, improved realisations and digital-led operating leverage. Zerodha, which is a key unlisted player in the discount broking space, said last year that they expect around 40 per cent impact on FY26 revenue due to tightening of the futures and options segment.

HDFC Securities reported a net profit of ₹268 crore and revenue of ₹850 crore in Q4, while full-year profit declined to ₹930 crore, due to a subdued first half and higher employee costs.

The brokerage continued to pivot towards a lean, digital-first model and grew its customer base by 2 lakh to 78 lakh in the previous quarter.

Kotak Securities, meanwhile, posted net profit of ₹400 crore for the quarter with stable full-year earnings, indicating resilience but limited growth amid competitive pressures and moderating trading intensity.

MIXED TREND

Smaller and mid-tier brokers showed a mixed trend. 5paisa Capital reported quarterly revenue of ₹85.5 crore and profit of ₹10.8 crore, though full-year profit declined sharply, due to pressure on margins and lower activity earlier in the year.

In contrast, Anand Rathi Share and Stock Brokers delivered strong growth, aided by a surge in margin trading funding, while SMC Global Securities reported a sharp jump in profits on a low base.

Industry-wide FY26 trends were shaped by fluctuations in retail participation, regulatory changes and a high base from previous years.

Analysts said the March quarter recovery was largely driven by higher market volatility, which typically boosts trading volumes, alongside improving client engagement.

However, the sustainability will depend on consistent inflows, product diversification and the ability to monetise a large but price-sensitive retail base.

No impact of family feud on record FY26 results, says Sona Comstar CEO

Amit Vijay Mohile
Mumbai

Vivek Vikram Singh, MD and Group CEO, Sona Comstar

Sona BLW Precision Forgings Ltd (Sona Comstar) posted record financial performance in FY26 and the March quarter, even as a dispute within the promoter family over control of the Sona Group — often pegged at around ₹30,000 crore — has escalated into a legal battle.

The Gurugram-based mobility technology firm reported Q4 revenue of ₹1,272 crore, up 47 per cent year-on-year (y-o-y), with EBITDA rising 32 per cent and net profit growing 17 per cent. For FY26, revenue rose 26 per cent to ₹4,475 crore, while EBITDA growth moderated to about 13.5 per cent. Battery electric vehicle (BEV) contribution touched a record 39 per cent in Q4.

The dispute between the family of late Chairman Sunjay Kapur, including his mother Rani Kapur and widow Priya Sachdev Kapur, centres on control of Aureus, the promoter entity that holds shareholding in the listed company.

CLEARING THE AIR

The legal battle has culminated with the Delhi High Court ordering status quo on the estate and the Supreme Court nudging mediation, suggesting investor focus remains on fundamentals.

Speaking to *businessline*, Managing Director and Group CEO Vivek Vikram Singh drew a clear distinction between ownership-level developments and business execution.

"There is zero impact on operations, on strategy, on governance, on how the company is run," Singh said, adding that the dispute sits "two levels above" the listed entity. He also pushed back against widely cited estimates of the dispute's scale. "There is a lot of confusion around the ₹30,000 crore number... I don't know how anybody came to that number," he said.

As the conversation shifted to business, Singh described FY26 as a year of recovery and acceleration. "Q4 was our best-ever quarter... after a terrible first quarter, everything seems like an upside," he said.

The rebound builds on a longer-term trajectory. "We've grown revenue about 6.4 times in seven years... despite Covid, semiconductor shortages and geopolitical disruptions," he added.

GROWTH DRIVERS

Singh outlined three clear drivers of growth: Electrification, new product ramp-ups and steady order inflows. "Our growth is coming from electrification, new products launched over the last two years, and new orders ramping up," he said.

Global trends continue to reinforce this trajectory, with EV adoption accelerating across markets. "Electrification is one of the few things that gains from disruption. Whenever supply chains are disrupted or oil prices rise, EV adoption accelerates — and that benefits us," he said.

Sona Comstar ended FY26 with a net order book of about ₹23,700 crore, representing a 10-year pipeline of expected revenue. The company operates across 67 EV programmes with 35 customers, and has diversified beyond passenger vehicles into two- and three-wheelers, commercial vehicles and non-automotive segments such as Railways.

'SIS eyes ₹20,000 cr revenue in FY27, accelerating market share, solutioning'

bl.interview

Shishir Sinha
New Delhi

An Indian multinational in the field of security solutions, facility management and cash logistics, the SIS Ltd aims to achieve ₹20,000 crore of business during the current fiscal after closing FY26 with an all-time high revenue of around ₹16,000. Group Managing Director Rituraj Sinha said that despite more usage of technology, especially artificial intelligence, opportunities for humans will not reduce.

Edited excerpts:

> SIS is a proxy for India's growth. If the economy is going to grow at 7-8 per cent, SIS will grow at 2x the GDP. This has been the trend for the last 20 years
>
> RITURAJ SINHA
> Group MD, SIS Ltd

What is the guidance for FY27?

SIS is focused on two metrics — 15 per cent growth and 15 per cent return on capital/return on equity. FY26 has been a year of rebound. We are exiting FY26 with the highest revenue for the quarter, highest margin, highest profit after tax. We believe that we have the momentum to make FY27 a stronger year.

What is the absolute number you are looking at?

We are working on ₹20,000 crore, but I think 15 per cent is better guidance for the market. SIS is a proxy for India's growth.

If the economy is going to grow at 7-8 per cent, SIS will grow at 2x the GDP. This has been the trend for the last 20 years. We have maintained a 15 per cent growth since 2018, when we were listed.

We believe that if ours is a compounding story, we need to execute 15 per cent growth and 15 per cent return on a compounding basis and the rest will fall in place.

The group has three divisions: security solution-India, faculty management and the security solution international. Which one will have better performance and share?

SIS Ltd clocks roughly ₹1,500 crore of revenue per month, as of March 2026. Out of this, roughly ₹850 crore comes from India and the remaining from international operations.

The Indian business is growing at roughly 15 per cent plus and we believe that this year, particularly FY27, should grow at 15 per cent plus.

We are the largest in India in security — almost double the size of the next competitor. We are also the largest in India in facility management.

We believe that now the focus has to be to grow the India businesses at a fast clip and accelerating market share growth and the migration from services to solution, which is basically adopting more technology with manpower, to give the customer solutions, not just the manpower service part. Accelerating market share and solutioning is our key focus in FY27.

What about the international business?

Since listing, the international business for the last 8 years has been growing at an average rate of 10 to 11 per cent per year. And we believe that it will maintain the same trajectory.

What would be the growth strategy — organic or inorganic? Any plan for acquisitions?

SIS is primarily a organic growth dependent company. We have made acquisitions as and when such opportunities have come up, which are for our shareholders, but growth is basically organic driven.

What are the new trends in the security solutions business?

Today, AI is being used very extensively in CCTV monitoring and remote surveillance, and these are adding a lot of value to our clients. AI for video analytics using CCTV footage is a mega trend, and we are using it.

The other mega trend is robotics. We were the first to bring robot dogs last year. Drones are also being used. SIS is using other devices for servicing and cleaning. Robotics is the second mega trend which we are very much focused on.

You have a manpower of 3.5 lakh. Now, with more use of AI and robotics, will we see the same growth in manpower recruitment in the coming years?

As of now, it's not a replacement technology, but a value-added technology. The guards and the janitors, cleaning staff are using these robots to do their services better. People in the CCTV control room are using AI video analytics to monitor better than watching the cameras themselves.

FPIs pull out ₹13,944 crore in holiday shortened week, equities hit hardest

Anupama Ghosh
Mumbai

Foreign portfolio investors (FPIs) remained net sellers in Indian markets for the shortened trading week ending April 30, pulling out ₹13,944.59 crore across equity, debt, hybrid and mutual fund instruments over four sessions — Monday through Thursday — as the markets stayed closed on Friday for Maharashtra Day.

According to data from the National Securities Depository Ltd (NSDL), FPIs recorded net equity outflows of ₹8,721.65 crore on Monday, making it the steepest single-session sell-off of the week. Tuesday saw net equity outflows of ₹4,188.34 crore, followed by ₹1,991.87 crore on Wednesday and ₹1,978.63 crore on Thursday — bringing the week's total net equity outflow to ₹16,880.49 crore.

DEBT MARKETS

While FPIs were net sellers in certain debt segments — particularly the fully accessible route (FAR) on Monday and Tuesday — they turned net buyers in FAR on Wednesday (₹1,338.95 crore) and Thursday (₹1,257.01 crore).

The voluntary retention route (VRR) segment also saw inflows on Monday (₹1,591.32 crore) and Tuesday (₹911.10 crore).

This came amid a cocktail of global headwinds. "During April, FPIs were sellers for ₹63,167 crore while they invested ₹2,319 crore through the primary market, taking net FPI outflows to ₹60,848 crore," said VK Vijayakumar, Chief Investment Strategist at Geojit Investments Ltd. He added, "The total FPI outflows from India in 2026 so far stand at a massive ₹1,91,968 crore."

Renewed tensions around the US-Iran situation and elevated crude oil prices weighed on risk appetite, with the rupee touching historic lows during the week.

DIIS CUSHION

Domestic institutional investors continued to absorb selling pressure, providing a floor to the market.

The Nifty 50 ended the week with a modest gain of 0.42 per cent, closing at 23,997.55, while the BSE Sensex edged up 0.33 per cent to close at 76,913.

K-food chain bb.q Chicken enters India with Bengaluru launch, eyes 150 outlets

Aishwarya Kumar
Bengaluru

South Korea's K-food wave is no longer limited to instant noodles and pop culture; it's now landing on Indian plates as global fried chicken chain bb.q (best of the best quality) Chicken makes its India debut.

Genesis bb.q group's flagship brand will open its first two outlets in Bengaluru in June, marking its entry into what it calls one of its most critical growth markets.

The rollout will quickly expand, with a Hyderabad outlet planned in July, followed by stores in Chennai, Kochi, Visakhapatnam and Vellore, taking the total to 7-8 outlets by the end of 2026.

The brand has partnered SrikG Group as its master franchisee for India, signalling a structured, long-term play rather than a limited-market test.

Globally, bb.q Chicken operates over 2,300 stores in South Korea and nearly 1,000 overseas, with the US emerging as its largest international market. The company is also entering Europe this year, underscoring its aggressive global expansion strategy.

FOCUS ON SOUTH

India, however, is central to its next phase. The company plans to scale to around 30 stores by 2027 and 150 outlets by 2031, with a near-term focus on south India before expanding into the north. Over the next four years, it is targeting more than 60 outlets across the southern markets.

Honggeun Yoon, Chairman of the Genesis BBQ Group, said India is among the company's most critical markets globally. "If there are three strongest countries in the world, it would be the US, China and India," he said, citing the country's large, young population as a key growth driver.

Duhyeon Yun (Chris), Global CEO, said India could contribute 15-20 per cent of the group's global business within five years, driven by a young, digitally savvy consumer base and growing exposure to global food trends.

QUICKLY.

OPEC+ clears small quota boost after UAE's exit

Major OPEC+ nations agreed to a modest and symbolic increase in June production quota levels, as the group sends a business-as-usual message following the surprise exit of the United Arab Emirates. Abu Dhabi, at the same moment, touted its own growth plans. Seven countries, led by Saudi Arabia and Russia, will add 1,88,000 barrels a day next month under the agreement, which was finalised at a video conference on Sunday, said OPEC. A small increase was expected by delegates before the UAE exit. The actual restoration of those barrels will depend on the Strait of Hormuz being reopened and shuttered production being restored. BLOOMBERG

AI to decide next CEO, cost cuts in May 7 meeting

New Delhi/ Mumbai: As headwinds persist for the loss-making Air India, its board is expected to discuss cost-saving plans, selection of the next CEO, financials and other issues during the meeting scheduled for May 7, according to sources. Headwinds for the Air India Group, which is projected to have incurred more than ₹22,000 crore loss in the financial year ended March 2026, have multiplied with the West Asia conflict. PTI

Foreign investors can set up insurance ventures in India, take over pvt firms

REGULATORY NOD. Aggregate holdings will be allowed in the automatic route, subject to IRDAI's approval

Shishir Sinha
New Delhi

With the latest amendment in the Foreign Exchange Management (Non-debt Instruments) Rules, 2019, to facilitate 100 per cent foreign direct investment (FDI) in the insurance business, except the Life Insurance Corporation of India (LIC), foreign insurers now have the option to either set up fully-owned ventures in India or take over private insurance companies.

According to the latest data, India has a total of 60 insurance companies, comprising 26 life insurance and 34 general insurance companies. Of the 26 life insurance companies, 25 are in the private sector.

PRIVATE SECTOR

Similarly, out of the 34 general companies, 28 are in the private sector.

Among the 53 private insurance companies (life and non-life together), four have foreign investment between 70 and 74 per cent, and the remaining between 18 and 49 per cent.

For a foreign insurer, an existing company in India is largely believed to be a better option considering the distribution network.

On Saturday, the Finance Ministry notified 100 per cent FDI in the insurance sector.

However, it will be capped at 20 per cent for the Life Insurance Corporation of India (LIC).

According to the notification, aggregate holdings by foreign investors, including portfolio investors, in the equity shares of an Indian insurance company are permitted up to 100 per cent of its paid-up equity capital.

It will be allowed in the automatic route, subject to approval and verification by the Insurance Regulatory and Development Authority of India (IRDAI).

RESIDENT CITIZENS

"In an Indian insurance company having foreign investment, at least one among the Chairperson of its Board, its Managing Director and Chief Executive Officer shall be resident Indian citizens," it said. Further, the foreign equity investment cap of 100 per cent will apply on the same terms to insurance brokers, re-insurance brokers, insurance consultants, corporate agents, third-party administrators, surveyors and loss assessors, managing general agents, insurance repositories and such other entities as may be notified by the IRDAI from time to time.

FDI proposals will be allowed under the automatic route, subject to verification by the IRDAI. The foreign investment in insurance intermediaries shall be governed by the Indian Insurance Companies (Foreign Investment) Rules, 2015.

If the intermediary is like a bank whose primary business is outside the insurance sector and is permitted by the IRDAI to function as an insurance intermediary, the foreign equity investment caps applicable in that sector will continue to apply.

It will be subject to the condition that the revenues of such entities from their primary (non-insurance-related) business must remain above 50 per cent of their total revenues in any financial year.

AMENDMENTS

In December 2025, Parliament passed the Sabka Bima Sabki Raksha (Amendment of Insurance Laws) Bill, 2025, which brought key changes to the insurance framework.

The legislation amended three major laws governing the sector — the Insurance Act, 1938; the Life Insurance Corporation Act, 1956; and the Insurance Regulatory and Development Authority Act, 1999.

Shareholding of private life insurance companies in India

Companies	Foreign		Companies	Foreign	
	Promoters	Investors		Promoters	Investors
Aditya Birla Sun Life	49	0	Generali Central Life	74	0
AGEAS Federal Life	70	0	HDFC Life	0	24.4
Aviva Life Insurance	74	0	ICICI Prudential	21.93	13.68
Axis Max Life	0	0	IndusInd Nippon Life	49	0
Bajaj Life	26	0	Pramerica Life	49	0
Bandhan Life	0	0	PNB Metlife	49.73	0
Bharti Axa	0	0	Shriram Life	23	7
Canara HSBC OBC	25.5	4.6	Star Union Dai-Chi	47.42	0
Edelweiss Life	18	0	Tata AIA	40	0

Source: Websites of companies (All figures in % / as on December 31, 2025)

Textile Ministry moves to shield clusters from W. Asia gas supply shocks

Amiti Sen
New Delhi

The Textile Ministry has launched an assessment of natural gas supplies across all textile and handicraft clusters in India to identify potential shortages.

The move comes as public sector energy corporation GAIL (India) Ltd has assured the Ministry that it can source gas from the spot market to cover disruptions in supplies to the sector, though probably at a higher cost to industrial units, sources said.

"GAIL has confirmed it is currently meeting at least 80 per cent of the natural gas requirements for these clusters, which are classified as a priority sector. However, in the event of further supply shocks and disruptions, the agency is prepared to procure from the spot market to meet demand. But the higher prices would likely be borne by the units," a source tracking the matter told *businessline*.

There are over 500 textiles and 700 handicrafts clusters in the country. Many units in the clusters depend on natural gas for processes, such as high temperature dyeing, and running their furnaces and machinery.

SEVERE DISRUPTION

Natural gas supplies in India have been constrained due to the severe disruption in global energy trade routes, primarily caused by the conflict in West Asia and the prolonged blockade of the Strait of Hormuz.

"India's total natural gas consumption is about 189 mmscmd, of which 97.5 mmscmd is produced domestically and the rest imported," per government figures. About 47.4 mmscmd, or 25 per cent of India's gas supplies, have been affected due to *force majeure* conditions (suspension of long-term supply contracts due to the ongoing war) invoked by the major Gulf-based producers.

India's textile and handicraft sectors, in clusters such as Surat and Sanganer, have been experiencing a severe shortage in natural gas and LPG, with many units facing partial or full shutdowns and workers migrating back to their hometowns.

HUGE DISTRESS

"The glass industry in Ferozepur, in particular, has been facing huge distress because of the shortage of natural gas. The government's focus is now on meeting the shortages faced by the units there so that livelihoods are not at risk," the source said.

The Textile Ministry's assessment of all textile and handicraft clusters, being carried out by an internal monitoring group, is aimed at avoiding more Ferozepur-like situations, the source added.

The government issued a Natural Gas Control Order on March 9, under the Essential Commodities Act, to manage gas supplies and protect the priority sectors.

Nestle India's focus remains on volume-led growth, says CMD

bl.interview

Meenakshi Verma Ambwani
New Delhi

> We are fortunate that from the top-end the quick-commerce piece is growing, on the other end, rural is growing — and both of them are growing much faster than the company
>
> **MANISH TIWARY**
> CMD, Nestle India

Nestle India posted strong volume growth in the March quarter across categories. The company has also been focusing on bolstering rural reach.

In fact, in the March quarter, India became the largest market for its global chocolate brand Kitkat.

Manish Tiwary, Chairman and Managing Director, Nestle India, speaks on a range of issues, including cost pressures arising out of the West Asia conflict, category growth drivers and the growing salience of rural markets.

Edited excerpts:

What helped Nestle India post strong volume growth in the March quarter?

This is not a one quarter phenomenon; we had reported strong growth even in the December quarter.

Even in Q2, we did not report any disruption despite the GST transition.

The government's incentives on taxation and the GST reforms worked well as macroeconomic tailwinds and there has been strong execution by the team.

What kind of cost pressures are you facing due to the West Asia conflict? Has it changed your strategy?

Last year, we had faced headwinds in two of our biggest commodities, cocoa and coffee, as the prices went through the roof.

That has softened now. Unfortunately, milk prices now to some extent, but more importantly, the rising packaging costs due to the West Asia conflict has emerged as headwinds. There are also concerns around predictions of below normal monsoon and its impact on demand.

While we are managing the cost escalations, what is more challenging is the unpredictability of it. But there is no change in our strategy.

We continue to focus on volume-led growth, driving cost efficiencies and leveraging technology while putting the consumer at the centre of it.

We will need to focus on disciplined execution and continue to invest behind brands and innovations.

At the beginning of the year, we had said that bulk of the growth will be volume-led. If inflation creeps, some of it could become pricing-led... But I still want it to be volume-led.

I mean, ideally post GST reforms and if the West Asia situation had not happened, it would have been one of those really awesome years.

Now, there are some question marks. But I'm still not writing off the year as it all depends on how long the West Asia crisis lasts.

We will need to watch out Q3 for West Asia and monsoon impact.

Have you started taking price hikes?

Almost 100 per cent of our manufacturing is done in India, and 97-98 per cent of our raw materials are also sourced locally.

We are seeing inflation on packaging material and utilities. Till now, we've absorbed it. How long can we absorb it? I'm not sure, and that's why we have to work harder to drive cost efficiencies. But till now, we have not passed anything on.

In the new fiscal year, which of the categories will be stronger growth drivers for Nestle India?

We have huge headroom for growth in all our categories as they have lower penetrations.

In terms of our established businesses, confectionery has been one of our fastest growing categories, while coffee and prepared dishes are pretty close to it in terms of growth rates.

Kitkat has emerged as one of our fast-growing brands, and India became the largest market for Kitkat in the March quarter ahead of the UK.

Coffee category has been doing consistently well for us over the past 10 years as we have been driving penetration and gaining market share.

Noodles has been picking up strong momentum over the last few quarters. All of this is volume-driven growth and gives us confidence.

Even the milk and nutrition segment has done much better in the last two quarters.

Meanwhile, pet food and Nestle Professional are growing much faster than the established categories as they are still small businesses.

The company has been focusing on growing rural distribution. How has this contributed to rural growth in sales?

So, if you go back five years, we were largely an urban company.

Then we went to smaller towns. In the last one-and-a-half years, we started going into rural.

We are strongly focused on growing our rural business.

In the last one year, we have grown our retail distributor wholesale hubs to 45,000 from 25,000.

We are ensuring that each of the 2,20,000-odd villages we go to are tech-enabled so that we have better control.

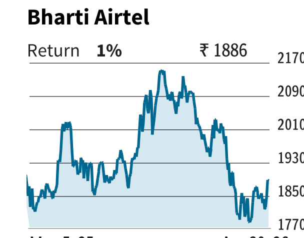
So, the rural contribution is between 22-27 per cent for us right now, but it is growing much faster.

We are fortunate that from the top-end, the quick-commerce piece is growing, on the other end, rural is growing — and both of them are growing much faster than the company. So, I'm equally excited by both the opportunities.

TODAY'S PICK.

Bharti Airtel (₹1,886.05): BUY

Akhil Nallamuthu
bl. research bureau



The stock of Bharti Airtel was in a downtrend in January and February. But the chart shows that in March, the bears lost momentum and the scrip formed a good base around ₹1,770. It has now rebounded on the back of this.

The price is now above both the 21- and 50-day moving averages and the stock recently surpassed a resistance at ₹1,870. In the coming sessions, there could be a price correction, possibly to ₹1,850. But then, we expect the stock to regain bullish momentum and eventually appreciate to ₹2,120 in the coming weeks.

So, traders can buy the stock now at ₹1,886 and accumulate at ₹1,850. Place the stop-loss at ₹1,800. When the price rises to ₹2,000 and ₹2,060, raise the stop-loss to ₹1,900 and ₹1,980, respectively. Book profits at ₹2,120.

Note: The recommendations are based on technical analysis. There is a risk of loss in trading

Listed realty firms tighten grip on land deals as market consolidates

T E Raja Simhan
Chennai

The real estate market is undergoing a structural consolidation, with listed real estate developers, such as Godrej Properties and Brigade, tightening their grip on prime land, even as overall transaction volumes soften. The listed players accounted for 49 per cent of all land deals in FY26, up from 40 per cent in FY25.

In absolute terms, 54 out of 111 land deals, covering over 1,433 acres, were executed by listed companies, highlighting their growing dominance despite a dip in overall deal volumes, according to data from the Anarock Group.

Leading the activity was Godrej Properties with 17 deals spanning 443.5 acres, followed by the Brigade Group with eight deals. Anuj Puri, Chairman, Anarock Group, said that land acquisition is increasingly becoming both capital-intensive and regulation-driven over the last few years.

'A CLEAR EDGE'

"In this scenario, listed developers have a clear edge over unorganised or smaller players, thanks to their easier access to institutional capital and transparent balance sheets. While the total number of land deals dropped from 143 in FY25 to 111 in FY26, the land-buying activity of these dominant players remained remarkably resilient," he said.

Ranjeeth Rathod, MD, DRA Homes, said the consolidation is structural. "Capital today is choosing credibility. Institutional capital is flowing disproportionately towards well-governed and scalable developers," he noted, adding that access to funding had become the biggest competitive advantage.

On solid ground

Total number of land deals	Deals of listed developers	% share of listed developers
111	54	49
Total land area (acres)	**By listed developers**	**% share of listed developers**
2,993.57	1,432.97	48

Source: Anarock Research & Advisory

> Leading the activity was Godrej Properties with 17 deals spanning 443.5 acres, followed by the Brigade Group with eight deals

TOP HOTSPOT

Rathod also pointed out that landowners were prioritising execution certainty over price, while regulatory tightening post-Real Estate (Regulation and Development) Act, 2016, had increased compliance costs, accelerating consolidation in favour of organised players.

Bengaluru emerged as the top hotspot with 17 deals, covering over 293 acres. Other key markets include Pune: 8 deals (78 acres); the Mumbai Metropolitan Region: 7 deals (51+ acres); Chennai: 5 deals (74+ acres); Hyderabad: 5 deals (38 acres) and the National Capital Region: 2 deals (18.6 acres), said Anarock.

Among emerging markets, Amritsar stood out with two deals totalling 520 acres, while Vadodara, Nagpur, Panipat, Mysuru, Raipur and Coimbatore also saw activity.

NEW HOUSING SUPPLY

Across the top seven cities, listed and Grade A developers accounted for 45 per cent of new housing supply in FY26, up from 43 per cent in FY25. The shift is especially pronounced in the National Capital Region, where 66 per cent of new supply came from organised players.

"This clearly highlights NCR homebuyers' rising prioritisation of reliability and brand equity... the market has undertaken a major flight to trust," said Puri.

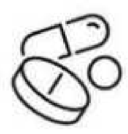
Gaurav Pandey, MD and CEO, Godrej Properties Ltd, reacting to the Anarock report, said: "We added 18 new projects in FY26 with an estimated saleable area of 33.32 million sq ft and booking value potential of ₹42,100 crore, more than double our initial guidance, making it our best-ever year for business development."

the hindu businessline.
MONDAY - MAY 4, 2026

Costly remedy

Sun-Organon deal, mitigating future market risk

The landmark $11.75 billion buyout of US-based Organon by Sun Pharma impacts investors, policymakers and consumers. At the outset, there is a looming context that cannot be overlooked. The Trump administration, after having concluded its Section 232 investigations on pharma imports recently (examining 'national security' concerns), decided last month that tariffs of up to 100 per cent will be imposed with effect from July, on some patented drugs and 'associated' ingredients.



Imports of generics will be exempted if the exporting companies enter into drug pricing agreements with the US, or if the country concerned gets into "MFN deals with the administration". This order really matters for Sun Pharma, which earns nearly a third of its revenues from the US market as an exporter of generics. With the latest US order, it would probably make sense for global companies reliant on the US market to cut deals to 'make in America'. The Sun-Organon deal might have been struck to mitigate future market and cost risks. A valuation of 6.2x of Organon's EBITDA looks good for Organon as well. Sun possibly gains by securing its US market and getting an entry into China (13 per cent of Organon's revenues). It can leverage Organon's product range, particularly in areas of women's health (33 per cent of its revenues), established brands (half its revenues) and biosimilars (11 per cent and rising sharply).

However, Sun is buying into Organon's debt, which is 4x its EBITDA, while its own debt-EBITDA ratio is below one. The challenge for Sun is to secure better long-term returns from Organon's products than the latter has managed in recent times. Since Organon has no major patents, it is not clear how Sun can leverage them. In its 2024 annual report, Organon has cited its "risks". Its sales are over-reliant on a few products, such as Nexplanon (its flagship product, a contraceptive implant), Arcoxir (a pain reliever for arthritis in particular) and Singulair (for respiratory ailments). Organon's sales from Nexplanon have dipped as its patent has lapsed in most countries except for the US, where the patent expires in 2027. According to the annual report, it will have to rely on "future acquisitions, partnerships or collaborations" even to keep branded generics sales going. Clearly, Organon lacks the means to carry on, even as its women's products and presence in biosimilars holds promise.

For investors and consumers, the combined entity (comprising two equals with revenue of $6.2 billion each) poses certain risks. A new set of products could enter the Indian market, amidst low levels of awareness. It is worth noting that Organon's CEO quit last year over sharp sales promotion practices with respect to Nexplanon and an anti-depressant, Remeron. The company faces trouble over alleged patent infringement, while its contraceptive NuvaRing is controversial for side effects. India should keep a strict regulatory eye on 'innovative' products, without coming under pressure.

OTHER VOICES.

The Guardian

When global shocks hit your shopping bill

When the Bank of England warned this week that food inflation could reach 7% by the end of the year, it revealed how little stands between a geopolitical jolt and a domestic crisis in Britain. A shock wave in the Gulf feeds through energy, fertiliser and supermarket prices into falling incomes, weak growth and job losses. What it exposes is not just inflation but a system unable to absorb disruption. The Bank is right that interest rates cannot move global energy prices. Raising them will not fix the shock. Instead, rate hikes redistribute the impact by compressing wages and deterring investment to stop higher costs becoming embedded. What appears as inflation is, in reality, the price of dependence on the strait of Hormuz. Clearly, the UK's stability rests on security that the country that has yet to build into its infrastructure. LONDON, MAY 1

讀賣新聞
THE YOMIURI SHIMBUN

Deepen Discussions in Light of Global Turmoil

The postwar international order based on multilateral cooperation and free trade is on the verge of collapse. People now live in an era where merely advocating for peace is no longer enough to protect their nation's security. It is the duty of politics to bring the supreme law in line with the times to tackle new challenges. It has been 79 years since the Constitution came into effect. Japan has enjoyed peace and successfully achieved economic development after the end of World War II. It goes without saying that the pacifist Constitution has served as the foundation for this. Universal principles — popular sovereignty, respect for fundamental human rights, and pacifism — must continue to be upheld. TOKYO, MAY 3

LINE& LENGTH.

TCA SRINIVASA RAGHAVAN

India's capital account obsession

India has not tried to bolster its current account. At the same time, it has chosen to squeeze capital flows, hurting growth

Why have successive Indian governments since the mid-1950s been obsessed with the capital account and ignored the current account almost altogether? This question is prompted by the statement by the Chief Economic Adviser on the record gross capital inflows in the last 12 months.

That's good news, doubtless, because more foreign money is good for the economy. But wait: when was the last time you heard any CEA talk about the current account, except in passing?

First, however, the definitions. The current account comprises a country's balance of trade, net of factor income and cash transfers. That means earnings from what it sells to foreigners. And the capital account shows how their capital is coming in and going out. It shows the country's overall financial position and the confidence foreigners have in the economy.

That out of the way, the answer to the question about CEAs and the current account was a dismaying nearly never. And this when the capital account is really the balancing factor in the balance of payments. It makes up the shortfall in the current account when a country exports less than it imports.

Before 1947 the capital account wasn't really an issue because the Indian economy was fully integrated with the British and therefore the global economy. Capital flows were strong and seamless. The Brits saw to it to ensure that they got the investments they wanted or needed. So India had full capital convertibility.

However, the current account did cause a lot of concern to our English lords and masters. They kept fiddling with the exchange rate to keep it in good shape, at least from their point of view. Starting from the early 1890s they made life difficult for us. You should read the controversies on and around the subject.

But after 1955 it's been the other way round. It is the capital account that governments have been worried about, not the current account even though what happens in the capital account is a consequence of what happens to the current account. Why has this happened? Basically what you don't earn, you borrow.

WHY THE CAPITAL ACCOUNT?

There can be many reasons. The most important reason has to do with the Second Five Year Plan and the foreign exchange requirement demanded for it by PC Mahalanobis. Such was the capital intensity of that Plan that it was assumed that the money would have to come into the capital account via loans and grants rather than via the current account as a result of earnings from exports. This when India's share of world trade was still above 2 per cent.

But the emphasis on capital intensive growth came at the expense of labour intensive investments into the export sector. We did the most stupid thing imaginable: we gave up our comparative advantage. You can almost hear Eli Heckscher and Bertil Ohlin groaning.

The capital goods model of industrialisation in the 1950s and the primacy given to the Finance Ministry over the Commerce Ministry are the reasons why the current account was ignored over the years

Another reason was administrative, namely, the division of work between the finance ministry and the foreign trade ministry, as it was then called. The former was in charge of the capital account while the latter, being in charge of exports and imports, was in charge, so to speak, of the current account.

Given the pecking order in government, the Finance Ministry and therefore the capital account took precedence in the setting of priorities. That hasn't changed since the mid-1950s. The Commerce Ministry has to take its chances on debt and exchange rate policies.

There are other reasons as well such as the political pressure on governments not to be seen as a debtor. The colonial image of a farmer drowning in debt was transferred to the country as a whole.

Pakistan has shown that this is false imagery. It's been to the IMF two dozen times. But Indian politicians, in their anxiety to avoid this, forgot that if you earn enough, you don't have to borrow. The current account was therefore ignored by them. In India, capital flows were constrained by fear of debt.

THE CONSEQUENCES

There have been two major consequences of this approach to the management of the external sector: one, a steadfast refusal to tap adequately into foreign savings even though our own savings are inadequate. Combine that with a refusal to turn around the current account, and we ended up with a capital shortage. The stigma attached to debt has impeded growth.

The second consequence is the sacrificing of capital efficiency because of import substitution. Not only do investments made behind tariff walls face reduced pressure to be efficient, there is a spillover effect on the labour market where wages are higher than they would otherwise be.

It's possible to enumerate many more such foolish things but the point ought to be clear. By focusing on the capital account and neglecting the current account, we have made a gigantic mistake. It can be rectified if the thinking changes.

MSP methodology should reflect changing reality

Refinements in interest cost and accounting for volatile inputs will help in encouraging cultivation of oilseeds and pulses

Mohit Sharma

India's Minimum Support Price (MSP) framework rests on a cost estimation system. At its core lies the Comprehensive Scheme for Studying Cost of Cultivation, implemented by the Directorate of Economics and Statistics.

The scheme relies on a triennial block sampling design in which selected villages are observed over a three-year cycle before rotation. Although methodologically sound, this framework has remained largely unchanged over decades, despite significant structural transformations in Indian agriculture.

The concern is that MSP recommendations are often based on cost conditions that are two to three years old. The 2021-22 input shock offers a clear example: global fertilizer prices surged, diesel prices increased before partial relief measures, and labour costs rose moderately. This indicates that the MSP system is structurally stable in normal periods but becomes systematically misaligned during input price shocks. There are evidences of research reports proving the gaps in actual plot level cost and MSP; had costs been more accurately captured, MSP for several crops would have needed to be 20–30 per cent higher to maintain the intended margin over costs.

A more subtle but increasingly important gap arises from changes in mechanisation patterns. The existing framework distinguishes between owned machinery (accounted for through depreciation and interest) and hired machinery services (captured under paid-out costs), which in principle avoids double counting. The Sub-Mission on Agricultural Mechanization (SMAM) has allocated a total of ₹8,565 crore across States for the 2014-15 to 2024-25 period, aimed at distributing over 1.9 million machines and establishing thousands of Custom Hiring Centres. This policy attention of the government on farm mechanisation, rapid expansion of custom hiring centres has shifted machinery access patterns, particularly among small and marginal farmers. The triennial sampling design, however, may still reflect older ownership-heavy distributions.

VARIED FACTORS

These methodological gaps influence the accuracy of cost estimation but do not, by themselves, explain broader structural outcomes such as cropping patterns. Evidence from across States suggests that crop diversification is driven far more by procurement assurance, irrigation conditions, input subsidies, and market linkages than by marginal differences in estimated costs.

States such as Rajasthan, Madhya Pradesh, Maharashtra have achieved relatively higher shares of pulses and oilseeds due to stronger market ecosystems, whereas Punjab continues to be dominated by the rice–wheat system.

SUPPORT PRICES. Need for tweaks

This contrast underscores a critical insight: MSP in practice operates less as a pure cost-based price signal and more as a procurement-backed assurance mechanism. Even where MSPs are announced for alternative crops such as pulses and oilseeds, weak procurement limits their effectiveness in shaping farmer decisions. Data shows that market prices for several crops have frequently fallen below MSP in multiple regions.

The implications for MSP outcomes are therefore twofold. First, during stable periods, the gap between MSP and actual cost remains limited, and the system functions broadly as intended. Second, during periods of input price volatility, lagged cost estimation can compress the MSP-to-Cost margins, reducing real profitability even when nominal MSP increases are announced.

The MSP debate in India is often framed as a pricing problem, whereas it is fundamentally a problem of measurement and transmission.

Addressing these issues requires calibrated reform. Refinements on interest rate assumption and introducing limited indexing for volatile inputs like fuel and fertilizers could be piloted for crops where diversification is desired, such as pulses and oilseeds. Later, these refinements could be extended across all MSP crops, accompanied by gradual improvements in sampling frequency and regional representation. The fiscal implications of such changes would remain modest relative to their potential to enhance policy credibility and precision.

India's MSP system has played a critical role in ensuring food security. Yet a framework designed for earlier production conditions now faces the risk of gradual misalignment as agricultural systems evolve. A carefully sequenced modernisation focused on improving cost estimation without disrupting institutional continuity can strengthen both its credibility and effectiveness. When cost estimates more closely reflect observed realities, MSP can function not only as a safety net but also as a reliable guide for long-term agricultural transformation.

The writer is Assistant Professor (Agri- Business), Dr. Rajendra Prasad Central Agricultural University (RPCAU), Pusa, Samastipur, Bihar. Views are personal

BELOW THE LINE

TVK leader Vijay during an election campaign BALACHANDAR L

Seeking Divine intervention

Joseph C. Vijay, founder of Tamilaga Vettri Kazhagam (TVK), embarked on a multi-faith spiritual tour ahead of the Assembly election results set to be announced on Monday, seeking divine blessings at key religious centres across the country.

He began his pilgrimage at the Tiruchendur Murugan Temple, where he offered prayers to Lord Murugan. Holding the *'Vel'* — the deity's sacred spear — Vijay performed 'darshan' at the coastal shrine before proceeding to the Shirdi Sai Baba Temple near Pune.

Continuing his tour, he was supposed to visit Shrine Basilica Vailankanni Church in Velankanni, a prominent Christian pilgrimage site, and conclude his Yatra at the Nagore Dargah, an important Islamic shrine but this got cancelled at the last minute even as a large number of his fans assembled at the Church waiting to see him. With several exit polls predicting a strong performance by the TVK, Vijay's outreach to multiple faiths underscores a carefully calibrated message of inclusivity, as he positions himself to emerge either as a 'king' or a potential 'kingmaker' in Tamil Nadu's evolving political landscape.

Golden opportunity?

In a recent event a retired SEBI official was all praise for himself for conceptualising an idea to facilitate the common man buy gold coins and bars on MCX in the most transparent market driven price. What one wonders is why should investors buy gold at futures prices when they are paying the entire money today with an intention to take delivery. Other participants use the exchange platform only for hedging their risk. Moreover, futures prices are always at a premium to spot prices as it involves cost of carry and storage. This trend is not going to change in the near future given the geopolitical turmoil.

Rudy's mid-air roll call

A routine IndiGo flight from Jammu to Delhi on May 1 had an unusual in-flight announcement: the pilot, Rajiv Pratap Rudy, casually welcomed Union ministers Ashwini Vaishnaw and Jitendra Singh, along with MPs Sanjay Jaiswal and Manoj Tiwari, turning a standard cockpit message into a mid-air political cameo.

A former civil aviation minister under Atal Bihari Vajpayee, Rudy has been repeatedly elected from Bihar's Saran Lok Sabha constituency. Though his political prominence has waned over the years, he has stuck with flying, operating Airbus A320 aircraft in an honorary capacity. That makes him a rarity: among India's politicians, only Rudy and the late Rajiv Gandhi have flown commercial passenger jets. Another famous airman, Rajesh Pilot, was an Indian Air Force officer, not a commercial airline pilot. **Our Bureaus**

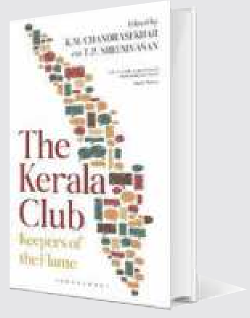
Published by Nirmala Lakshman at Kasturi Buildings, 859-860, Anna Salai, Chennai-600002 on behalf of THG PUBLISHING PVT LTD., and Printed by Ra Babu Vijay at THG PUBLISHING PVT.LTD. Plot B-6 & B-7, CMDA Industrial Complex, Maraimalai Nagar, Chengalpattu Taluk, Chengalpattu Dist., Pin: 603209. Editor: Raghuvir Srinivasan (Editor responsible for selection of news under the PRP Act). ISSN 0971 - 7528

Seven decades of Keralam

29 bureaucrats offer a rich portrait of Kerala

BOOK REVIEW.

S Adikesavan



Title: The Kerala Club: Keepers of the Flame
Edited by KM Chandrasekhar and TP Sreenivasan
Publisher: Bloomsbury
Price: ₹799

Kerala consistently punches above its weight in any discussion on social and economic development in India. Its achievements in human development indicators, as well as its distinctive administrative challenges, have long attracted national attention. Further, until about a decade ago, key positions in the Government of India —particularly in the Prime Minister's Office — were occupied by officers of Malayali origin.

The phrase "Mallu Club" or "Keralite Mafia" came into currency to describe this influential cohort during the tenure of Manmohan Singh. Officers such as TKA Nair (Principal Secretary to the Prime Minister), MK Narayanan (National Security Advisor), and Shivshankar Menon (Foreign Secretary) formed part of a powerful administrative axis in New Delhi.

The Kerala Club: Keepers of the Flame, edited by KM Chandrasekhar and TP Sreenivasan, (Bloomsbury, 2026) both guide and Guru to scores of young civil servants, brings together essays by 29 bureaucrats who either belonged to the Kerala cadre or had significant professional engagements with the State. The volume is both a chronicle and an analytical exploration, illuminating Kerala's developmental journey over roughly seven decades.

Several essays stand out for their narrative richness, personal touch and insight. In "A Heady Mix of Maths, Science and the Baton," Rajan Medhekar recounts his remarkable career trajectory — from a strong academic foundation in solid-state physics to leadership roles in the State police, the Border Security Force and as chief of the National Security Guard's "Black Cat" commandos. His reflections include a vivid account of the Chalai riots of 1982, one of the rare instances of communal unrest in Thiruvananthapuram, and the swift deployment of the Army to restore normalcy.

LABOUR RELATIONS

A recurring theme in this volume is the complexity of labour relations in Kerala. In "Kerala's Trade Unions," EK Bharat Bhushan, former Chief Secretary, offers a candid account of his tenure as Managing Director of Kerala Minerals and Metals Ltd. He describes how efforts to revive the company — through stricter discipline and improved marketing — were met with resistance from entrenched union leadership. The backlash included threats and even a legislative motion seeking his removal.

Transferred thereafter to Travancore Titanium Products, he encountered an even more fragmented union environment, with 48 (sic) unions operating simultaneously and discipline virtually absent. His essay is essential reading for anyone seeking to understand Kerala's industrial climate.

Equally compelling is "When the Desert Bloomed" by T Balakrishnan, which critiques the State's fiscal priorities. He argues that tax revenues have often been directed more toward welfare expenditure than toward critical physical infrastructure. Successive governments, he notes, have competed to enhance salaries and benefits for organised labour across both public and private sectors, contributing to wage inflation and rendering labour increasingly unaffordable.

The book will appeal not only to general readers interested in the state's development trajectory but also to serving bureaucrats and civil service aspirants.

That said, the volume might have benefited from a concluding synthesis drawing together key insights and offering a critical overview in summary. The diversity of styles and topics occasionally makes the reading uneven. Yet, this is a minor drawback in an otherwise valuable compendium.

The reviewer is a commentator on banking and finance

A fearless activist and a rebel

This compelling biography describes Annie Besant's many transformations and her role in India's freedom movement

BOOK REVIEW.

Parimala S Rao

Any guesses who the first woman president of the Indian National Congress was? She presided over its Calcutta session in 1917 and was, in fact, British national, political firebrand and supporter of Irish self-rule, Annie Besant. And this is just one of the many avatars that Clare Paterson describes so vividly in *The Nine Lives of Annie Besant*, the recent biography of a key force behind India's Home Rule movement and a dominant figure in the Theosophical Society, which celebrated its 150th anniversary last year.

The engrossing life story brings alive one of the most colourful figures of the late 19th and early 20th centuries — an inspirational activist and educator who left her imprint on many schools and colleges, localities and roads across India that are named after her. In recording Annie Besant's remarkable journey across continents and ideologies, Paterson presents a study in political and spiritual transformation. She shows how Besant's early activism in Britain laid the intellectual and moral foundations for her later, pivotal role in India's struggle against British rule.

Born in 1847, Annie, as she was popularly known in London's East End, was a revolutionary thinker, swept up on an extraordinary roller-coaster of a life. Losing her father when she was five, she and her older brother were brought up by their mother in a Victorian middle-class family. An intelligent, intuitive and deeply religious young girl, Annie was riveted by tales of the sufferings of early Christian martyrs, writes Paterson. Ironically, this unshakeable faith was shattered after her disastrous marriage, at age 20, to clergyman Frank Besant. Parting ways from her husband of six years and forced to give up their son to his care, Annie turned breadwinner for her mother and daughter with short-term jobs.

As a prolific writer, she was drawn to unconventional thinkers of the time, and became a pamphleteer in the early 1870s.

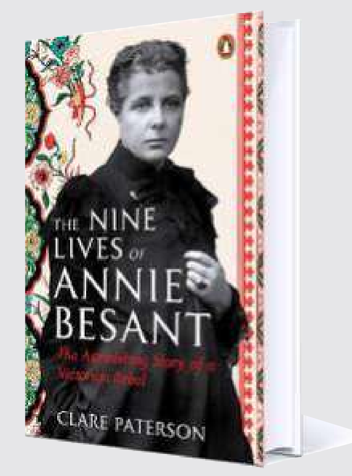
She wrote prodigiously, on vital current issues such as women's suffrage, unemployment, workers' rights, corruption, slavery, political reform, the lunacy laws, and much more.

MULTIFACETED PERSONALITY

Being a powerful orator, Annie was also a sought-after public speaker, captivating large crowds with her impassioned campaigns for birth control, workers' rights, and women's emancipation. Paterson highlights her involvement in freethought circles, her collaboration with politician, radical thinker and proponent of atheism, Charles Bradlaugh, and her willingness to court controversy while defending principle. The legal action and trial relating to her pamphlet on birth control underscore Annie's courage in challenging state and church authority, and her faith in fearless public engagement.

Annie's early radicalism, Paterson argues, did not disappear with her later turn toward spirituality; it evolved. Describing her meeting with Helena Petrovna Blavatsky in 1889, Paterson writes, "Madame Blavatsky's appearance was ramshackle but her personality was electrifying. She smoked incessantly and swore prodigiously. She rejected all accepted norms, from the church to science... It must have been invigorating for Annie to find so powerful a woman at the head of this organisation."

Annie's embrace of the Theosophical Society offered a broader framework that accommodated both spiritual inquiry and social reform. This new philosophy's emphasis on universal brotherhood and Eastern wisdom allowed multiple trajectories to converge, and resonated with her long-standing critique of Western imperial arrogance. She believed that self-government, or Home Rule, required not only political change but also moral, intellectual and cultural renewal. The account of Annie's role in India's Independence movement forms the political core of the book.



Title: The Nine Lives of Annie Besant
Author: Clare Paterson
Publisher: Penguin Random House India
Price: ₹499

ABOUT THE AUTHOR

Clare Paterson is an award-winning television executive with extensive experience of commissioning and making documentaries, some of which have been honoured with BAFTAs and International Emmys

FREEDOM FIGHTER

Paterson captures both the scale of her influence and the tensions inherent in her position as a British-born advocate of Indian self-government, which led to her imprisonment for political activism. When Annie assumed presidency of the Indian National Congress in 1917, Paterson says, she acquired "a huge base among nationalists as well as Theosophists, the sort of support that is every politician's dream."

British officials in India and England didn't know what to make of her but understood that "this formidable woman now at the centre of the imperial picture was refusing to toe the government line," while occupying "an extraordinary position of power as president of both a new world religion and of the mainspring of Indian nationalism."

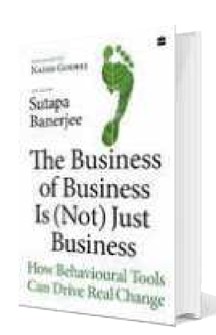
Paterson recognises the contradictions in Annie's radical politics — her preference for constitutional methods, her unease with civil disobedience, and her eventual marginalisation as nationalist strategies shifted. The book leaves the reader viewing Annie as a bridge between Britain and India, rationalism and mysticism, dissent and institution-building.

The Nine Lives of Annie Besant stands out for the clarity and edgy elegance of Paterson's prose.

She balances colourful narrative with analytical depth, enlivening the text with delightful and wondrous anecdotes. The result is a biography that is both intellectually rigorous and hugely readable.

The reviewer is an independent editor based in Besant Nagar, Chennai

NEW READS.

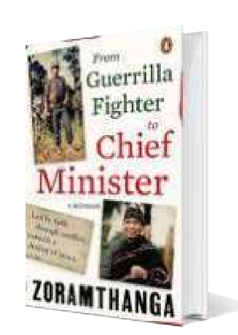


Title: Business Of Business Is (Not) Just Business

Editor: Sutapa Banerjee

Publisher: HarperCollins India

This book examines what it takes to move from box-ticking to meaningful action, and sync success with public purpose

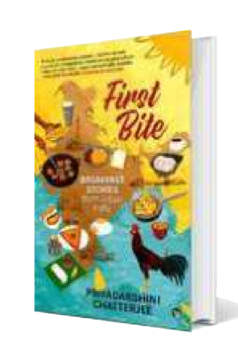


Title: From Guerrilla Fighter to Chief Minister

Author: Zoramthanga

Publisher: Penguin Eight

This is not just a political memoir. It is a story of struggle. A story of conviction. A story of one man who held his people's future in his hands



Title: First Bite: Breakfast Stories from Urban India

Author: Priyadarshini Chatterjee

Publisher: Speaking Tiger Books

This book blends food history with cultural reportage. It is a book about ordinary meals and extraordinary stories

thehindu businessline.

TWENTY YEARS AGO TODAY.

May 4, 2006

Petro price hike likely by third week

The expected increase in petrol, diesel and cooking gas (liquefied petroleum gas) prices may take place by the third week of this month. The Secretary in the Ministry of Petroleum and Natural Gas, Mr M.S. Srinivasan, told select media persons here on Wednesday that a final decision on price revision would be taken after consultations be tween the Petroleum and Finance Ministries.

ADB plans to double hike to India

The Asian Development Bank (ADB) has decided to more than double its financial assistance to India to over $2.6 billion by 2008 from the current level of around $1.3 billion, according to the ADB President, Mr Haruhiko Kuroda.

Gujarat Ambuja to merge Ambuja Cement with itself

Gujarat Ambuja Cements Ltd will amalgamate Ambuja Cement Eastern Ltd with itself, offering four equity shares of GACL of Rs 2 each for every five equity shares of ACEL of face value of Rs 10 each. This share ratio recommended by the GACL board on Wednesday (and also by the ACEL board), was based on a valuation report prepared by Deloitte Haskins & Sells, said a statement from the Ambuja Group.

Short take

Working towards a sensible AI plan in telecom

SP Kochhar

Recent geopolitical conflicts have demonstrated that telecom systems are no longer neutral utilities but active arenas of strategic competition. In this context, the concept of "sensible AI" becomes critical.

During the recent Iran war, cyber operations targeted communication systems, disrupted infrastructure and manipulated information channels before and alongside physical actions. This marked a shift from the isolated cyberattacks to coordinated, multi-domain strategies.

Telecom networks were used not only to disrupt enemy coordination but also to influence perception. By exploiting mobile applications and communication platforms, actors could send misleading messages, spread disinformation and create confusion among both civilians and military personnel. This highlights a critical reality: telecom networks are not just carriers of information — they shape it. In such volatile scenarios, artificial intelligence is increasingly used to monitor, manage and defend telecom systems. However, not all AI is equally effective. What is required is sensible AI. AI that operates with context awareness, restraint and reliability.

A sensible AI system does not simply react to anomalies; it interprets them. For instance, a sudden spike in network traffic during a conflict could indicate a cyberattack, or it could be a surge in legitimate communication. An overly aggressive AI might shut down parts of the network, causing unnecessary disruption. A sensible AI, on the other hand, evaluates multiple signals before acting, ensuring stability while addressing real threats.

Sensible AI in telecom must therefore evolve beyond conventional cybersecurity. It must integrate awareness of operational technology (OT) systems and understand how digital commands translate into physical outcomes.

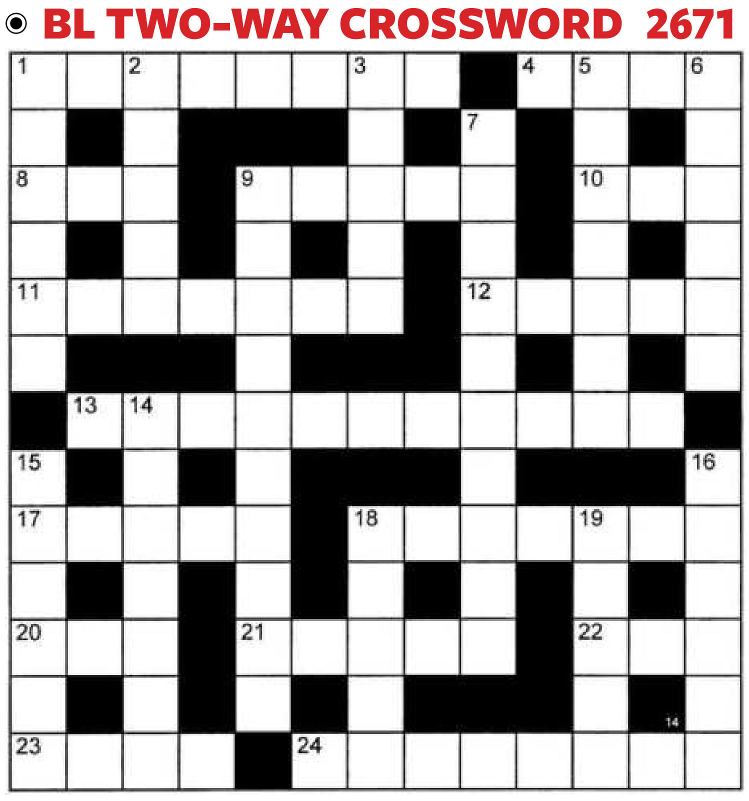
Another key takeaway is the importance of resilience. In conflict scenarios, network disruption is often inevitable. The goal is not to prevent all failures but to ensure continuity of critical services. Sensible AI enables the development of self-healing networks — systems that can detect faults, isolate compromised components and reroute traffic dynamically.

Modern telecom systems must move from reactive to predictive models. Instead of responding to attacks after they occur, sensible AI should anticipate potential threats based on behavioural patterns and historical data.

Sensible AI represents a shift toward systems that are not only intelligent but also measured, context-aware and resilient.

The writer is Director General, Cellular Operators Association of India (COAI)

BL TWO-WAY CROSSWORD 2671



EASY

ACROSS
01. An invective harangue (8)
04. Biting, sour (4)
08. Stud, plate in sole of shoe (3)
09. Dressing poured over food (5)
10. Russian village community (3)
11. Orange-coloured fruit (7)
12. Pungent bulb (5)
13. Beheaded (11)
17. Tree emblematic to Canada (5)
18. Escort vessel (7)
20. Practice, operation (obs) (3)
21. Two, couple, pair (5)
22. Sail of windmill (3)
23. Work set or undertaken (4)
24. Form of ninepins (8)

DOWN
01. Appal, discourage (6)
02. Carpenter's bore (5)
03. Not sharp (5)
05. Draw up, compose (7)
06. Throughout the time of (6)
07. Winding-up part of a speech (10)
09. Religious mysteries (10)
14. Fast train (7)
15. Quantity (6)
16. Court game (6)
18. Open, candid (5)
19. Smith's iron block (5)

NOT SO EASY

ACROSS
01. Help to turn the people with a bitter harangue (8)
04. Returning some medication that tastes sour (4)
08. Some of these guys have a stud in the shoe (3)
09. Dressing for dinner? What cheek! (5)
10. I'm backing the first Russian space station (3)
11. Fruit from a tropic being distributed (7)
12. There's no backing one on this sort of dome (5)
13. Having lost head when taped, one acted strangely (11)
17. Tree, the palm, put out at end of grove (5)
18. Vessel with its set of sails in its appointed lot (7)
20. Old-fashioned use of river in Yorkshire (3)
21. Mark Clemens made for the two of them (5)
22. Commercial vehicle used to test ore (3)
23. Piece of work terrorist leader will request (4)
24. A parody, lest it be confused with ninepins (8)

DOWN
01. The underworld is allowed to discourage one (6)
02. Argue about how boring it can be (5)
03. Take the edge off it by being outspoken (5)
05. Police about to take in first murderer to put it together (7)
06. While it's going on run, dig round it (6)
07. For each a speech, with this to wind it up (10)
09. Religious rites meant crass conversion (10)
14. A daily will put it into words (7)
15. How much for a horse to ride? (6)
16. Nine involved in street up at Wimbledon maybe (6)
18. This old German seemed sincere (5)
19. Smith strikes one villain perhaps abandoned by the Italian (5)

SOLUTION: BL TWO-WAY CROSSWORD 2670

ACROSS 1. Digestion 5. Sup 7. Cash 8. Deprived 10. Mushroom 11. Oral 13. Labour 15. Fallow 18. Doom 19. Ruminant 22. Commands 23. Once 24. Tar 25. Digresses

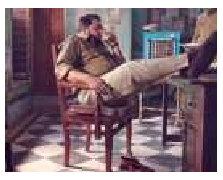
DOWN 1. Decimal 2. Gusts 3. Ice-box 4. Norm 5 Several 6. Pedal 9. Fraud 12. Ratio 14. Bloomer 16. Witness 17. Gun-dog 18. Ducat 20. Annas 21. Ward

CHENNAI
businessline.
MONDAY - MAY 4 - 2026

ON THE CAMPAIGN TRAIL.

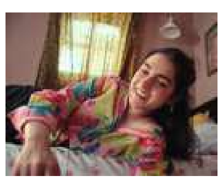
Even before we could step into May, brands were already pulling out giftable ideas for Mother's Day. Ad genius Piyush Pandey's final campaign for Fevicol was unveiled by the brand

The hot seat



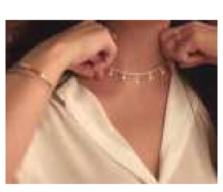
Pidilite Industries' latest Fevicol television commercial, 'Kursi Pe Nazar', captures how the humble chair transcends its role from just a furniture piece to becoming a symbol of aspiration and authority across homes, offices and institutions. The film shows how, in Indian society, who sits on a chair, who waits for it and who eyes it often signal shifting equations of ambition, influence and progression. Significantly, 'Kursi Pe Nazar' is the final Fevicol campaign ideated by the late advertising legend Piyush Pandey — the core thought being his, and now executed by his team. The film reinforces the brand's philosophy of strong bonds through a uniquely Indian behavioural insight.

Endless choice



Amazon India's new campaign for its 'Aur Dikhao' platform, brought to life by WPP, reinforces the e-tailer's position as the go-to destination for endless choice, trend-forward selection and great value for everyday shopping needs. Built on the simple yet deeply human insight that consumers rarely settle at the first option and instinctively ask for more, 'Aur Dikhao' transforms a familiar phrase into a powerful commerce behaviour platform. It captures the joy of discovery, the desire for better choices and the expectation of finding style and value within budget. Reimagined for a new generation of shoppers, the campaign launches across Hindi, Tamil, Telugu, Kannada, Marathi and Bengali with culturally rooted storytelling tailored to suit regional audiences.

The emotion of gifting



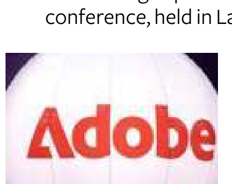
CaratLane's Mother's Day campaign featuring Yami Gautam Dhar focuses on the emotion of gifting. The campaign introduces a dual-gifting idea — buying a diamond for one's mother while also purchasing something for oneself. The campaign, which feels personal and relatable, focuses on occasion-led jewellery purchases, positioning Mother's Day as a moment for both gifting and self-purchase.

First at the table

In most households, mothers are usually the last to sit down for meals. But is it always out of sacrifice? Could it simply be because she craves that one quiet moment to savour her meals after tending to everyone else in the family. This Mother's Day, Badshah Masala set out to gently challenge this routine. Conceptualised and executed by Tonic Worldwide, the 'Maa First' campaign has popular influencers such as Seema Marathe, Ansh Dhote and Sonia Kotak's daughter recreating their mom's favourite dishes and serving them first — flipping the script to show just how much she is valued.

NUGGET.

Adobe's brand visibility tools for an AI-powered world



At the recent Adobe Summit, the tech leader's flagship customer experience conference, held in Las Vegas, the company announced a brand visibility solution for an AI-driven future. AI-powered chat services and browsers have increasingly become a primary channel for how consumers discover brands, evaluate and act. There has been a 269 per cent year-on-year increase in AI-driven traffic.

Adobe said its solution works as a continuous operating model across four vectors — sense, generate, reach and learn. Brands sense how they appear across AI-driven discovery surfaces, generate content and experiences grounded in brand context, reach both human audiences and AI systems from a shared foundation, and learn from every interaction to improve performance. Powered by AI agent and human collaboration, the advantages strengthen with every cycle.

"There is a new intermediary between brands and their customers, and unlike every one that came before it, this has the ability to reason," explained Loni Stark, Vice-President of Strategy and Product, Adobe. "For decades, brands have managed content, but now they also need to manage context to pinpoint what AI understands about their offerings and the institutional knowledge their own agents need to act — challenges that can be solved with our new solution."

Adobe Experience Manager adds a layer for managing the brand truth, permissions, governance and content sources.

◉ DIGITAL BLUEPRINT

Ikea's omni-channel retail transformation

MODERN ASSEMBLY. Parag Parekh on the Swedish retailer's app-driven experiments in India that it will take to the rest of the world

Meenakshi Verma Ambwani

Ikea is helping customers reimagine and visualise living spaces through AI. Parag Parekh, Global Chief Digital Officer of Ingka Group, the largest Ikea retailer, spoke with *businessline* during his recent India visit on the company's digital investments in India and how AI is reshaping consumer experiences.

Edited excerpts:


ILLUSTRATION: PARTHA PRATIM SHARMA

AI IN DIY. Parag Parekh, Global Chief Digital Officer, Ingka Group

How is AI transforming the retail industry and how is Ikea leveraging it?

AI is one of the most transformational technologies that the world is seeing. More importantly, the technologists themselves are surprised at the pace at which it is developing, especially since governments and corporations need to catch up sooner with the pace of the development.

When I joined Ikea about five years ago, a part of my assignment was to digitise Ikea. Our focus was on how to go from being a big box, physical brick-and-mortar retailer to an omni-channel retailer. I think we have come a long way now, with 30 per cent of our business being omni-channel, digital included.

That also meant we had to adjust our business model and build the muscle for some of these services that we were not used to even 10 years ago.

We have been on this transformation journey from physical to omni-retail. And while we are doing that we need to be ready for a transformation from an AI perspective.

We are working on three different aspects of transformation within the organisation. The first is around customer experience. I believe AI will fundamentally change how customer experience will look like in the digital world. As we are all used to websites and apps, I believe there is also quite a bit of friction in this. But AI makes it easier and conversational. And if you think about it, that will fundamentally change how e-commerce would look like.

And we are working with OpenAI and Google to build a customer experience that also incorporates AI and what it means in terms of the web, app or agentic commerce of tomorrow. So we are working through that.

Also, we are looking at the areas in our back-office that we can digitise to drive productivity and ensure more man hours are spent on customer experience. We are also working on integrating AI in our supply chains for improvement in areas such as forecasting accuracy and order allocations, among others, so our products reach the customer faster and at a lower cost in a sustainable manner.

Of course, in all of this, we will always ensure that customer trust is at the heart of the transformation. At the core of Ikea's digital transformation lies a commitment to ethical and responsible AI. We have clear policies on where we will use AI, where we will not use AI and how we will put AI in the service of our customers and co-workers.

As you expand omni-channel presence, have you begun deploying some of these AI capabilities in India?

We will use the AI capabilities more in India as it is a very app-driven market, rather than a web driven market. In many cases, we will experiment with the app in India and see how we can bring it to the rest of the world as well.

We have also doubled down on India by setting up the digital hub in Bengaluru. This team caters to building products not just for India but also our global operations.

We also believe India has a fantastic start-up ecosystem. Last year we acquired a leading logistics company, Locus, which will now power last-mile operations for India and globally as well. (The company, while based in the US, operates mostly out of India). We are continuing our omni-channel expansion in India. While the customer's journey starts online, we believe the store is where the centre is, as it enables consumers to get the full, immersive experience of Ikea.

You talked about the customer journey changing with the advancement of AI. How is that change taking shape at Ikea?

We have built AI-based design capabilities that our co-workers use internally to plan for customers. So, for instance, if earlier it took our co-workers 6-7 hours to plan one room for a consumer, it has come down to about 30 minutes now and we have improved the scale of quality.

We have also taken the next step to put this technology in the hands of consumers through Ikea Kreativ, which is an AI-driven digital design experience. It allows consumers to scan a room or any kind of living space and visualise how various Ikea products fit in with their creative vision.

We are working on experiences that offer consumers inspiration with 3-4 versions of designs for a living space at different price levels, rather than them having to start from scratch.

Then we allow you to take that journey forward from there or even fast-forward it by using conversational commerce.

So, it is really putting our expertise at the service of the customer through AI, conversational commerce and creative planning.

How are you ramping up digital investments?

We firmly believe digital will be one of the cornerstones and that is a strong commitment from our board to continue to invest in our digital operations that enable us to grow and scale up. So we will be doubling down on investments on the India digital hub and continue to grow it.

◉ PIVOTS AND TURNS

Steering a new course, wearing a little magic and Tvarra

Chitra Narayanan

Pivots have transformed the fates of several companies. It certainly saved Alpana Parida's helmet start-up from a big crash. Parida, a well-known marketing professional with stints at Singer, Tanishq and Future Brands' design firm DY Works (which she led), plunged into entrepreneurship some six years ago with a company called Tiivra Ventures to produce high-performance, composite fibre helmets for serious bike riders. But the making cost was too steep to sustain sales in the niche segment. She then pivoted to producing stylish and safe helmets for women and children riding pillion.

"If I had not already finished it, my book would definitely have had this case in it," exclaims Tata Play CEO Harit Nagpal, an early investor in Parida's startup and the author of *Pivot*. He says he knew little about helmets but knew Alpana well and invested in the person, rather than the idea. "If you invest in the person, you know they will steer the company well or, when it hits an obstacle, re-steer it," he says. Parida confesses that in December 2024 she was staring at failure until her focus pivoted to the pillion rider.

FROM TIIVRA TO TVARRA

"Every time I went into a helmet shop, I would wonder where was the women's helmet? Even when available they would be made from the men's moulds — the shapes were not right," she says. She thought of focusing on women's helmet but realised the market was small, with just 20-30 lakh women two-wheeler owners. "Then I thought of the pillion rider and the market suddenly became 30 crore plus," she says.

The catch, however, was that most pillion riders were buying cheap helmets just to stay within the law, not focusing on safety. "This was a big market, a white space and a problem worth solving," she says. Creating a new brand, Tvarra, Parida set out to unlock the market through design and marketing innovations.

To start with, Tvarra helmets are like fashion accessories — in vibrant colours with eye-catching designs and floral motifs. They are convenient, aesthetic and safe. She solved for common problems women face when wearing helmets — such as providing notches for earrings, which otherwise get jammed in a regular helmet. A range of accessories, such as matching arm sleeves, funky coloured helmet locks, scarves or balaclavas to keep hair unruffled (soon to be launched) show that beauty and safety can be companions. There's a tiny pocket within the helmet for an 'intention card'. "We ask women to write their intention and ride with it," says Parida.

TURNING HEADS. Alpana Parida, founder of designer helmet brand Tvarra

Incidentally, Tvarra's logo has two butterflies (the two Rs are shaped like wings) to denote someone taking flight. "Because if you think of the moment a girl sits on a scooter and the agency she has, it's that moment of genuinely taking flight," explains Parida. The tagline "wear a little magic" is pushed home through compelling storytelling.

CONTENT, COMMUNITY

Parida says she has absorbed many lessons from the people she has worked or studied with. From Falguni Nayar, her IIM-Ahmedabad classmate, she learnt that content and community are crucial to building a brand. Incidentally, it was Parida who coined the brand name Nykaa. "Content and community is what built Nykaa. That's what I am doing at Tvarra too," she says.

From CaratLane and its Founder Mithun Sacheti (he is an investor in Parida's startup), she has imbibed a collection-based approach. "The fact that in the jewellery category, where there are huge numbers of products, CaratLane broke it down with a collection-based approach, creating stores with minimal products but lots of storytelling, was inspiring," she says. So Tvarra helmets are grouped into collections with attractive names like Dreamcatcher, Blossom, Lotus, Daisy, Lilt and Flora.

While the D2C brand relies on online channels, including quick commerce, Parida has thought creatively about offline distribution.

The kids' helmets are distributed at bookstores (Crossword) rather than scooter showrooms to catch the eye of parents shopping with their kids. The storytelling and positioning is one where men will gift the women and children in their life the helmet to keep them safe. The brand film echoes this, heroing such men. With cricketer Jemimah Rodrigues recently padding up for Tvarra as investor-partner, Parida's venture is now on the fast lane.

Gen-Zers and their extraordinary new/old pastimes

Young people seek connection, growth and a respite from endless digital madness

THE CURIOUS MARKETER

HARISH BHAT

Earlier this month, I was speaking to a young man in his 20s about his interests outside work. He told me that the pastime that mattered most to him was playing board games. This surprised me because he is a digitally savvy, AI-native sort of guy. And yet, on weekends he heads to a board games cafe.

I asked him why, and this was his reply: "When I play board games, which require strategic thinking, I am doing something productive. Also, I am engaging with other people and friends. We keep chatting as we play."

Let me talk about another interaction that surprised me. One Saturday evening, I was walking on a busy road in Bengaluru when two young women stopped to say 'hello'. They knew me from a lecture I had delivered recently. I asked them what they were doing in that part of town.

One of them replied, "We have just attended a meeting of our book club." Two young women, Saturday evening, and a book club? Why, I wanted to know. "When we discuss a new book together, we get so many nice thoughts and ideas. We feel much better too. And we also get to know so many more people!" she replied.

'GRANDMA HOBBIES'

Both these encounters set me thinking. Our cities have many digital gaming parlours. We have apps that can stream into our bedrooms all the shows we may want to see. We have social media platforms with limitless content. And yet, it is the likes of board games and book clubs that are becoming sexy once again, especially to a sizeable segment of 20-somethings. Clearly, the times they are a-changin'.

I explored further and discovered other popular Gen-Z hobbies across India. These include running-and-breakfast clubs, supper clubs, which are also described as social mixers, baking cakes, and crafts such as embroidery and knitting. Some of these are pastimes we associate with our grandmothers, but they are back with a vengeance, trending among youngsters!

What are the fundamental reasons underlying this phenomenon? Here is my initial hypothesis.

Respite from digital overload

Young people (and older too) are confronted with digital screens everywhere — mobile phones, laptop computers, connected televisions, et al. They are also drawn into endless and addictive doomscrolling on social media. Among many Gen-Zers, there is a strong urge to break away from digital addiction. Pastimes such as knitting or baking, which occupy many hours at a time, are a good way to keep mobile phones aside for long durations.

GATHER AROUND. Board games have their digital-native fans VELANKANNI RAJ B

Connection with people

There is a strong desire among youngsters to connect with real people. Social media mostly offers superficial connection on the back of likes and comments, and overuse tends to make one lonely and depressed. For a generation that has grown up in this digi-gloomy environment, pastimes that involve sitting together with other people and talking to each other, such as book clubs or board games, offer great human connection. In fact, you meet not just friends but also other people at these places, which can potentially pave the way for new friendships.

Sense of purpose

Board games of strategy can develop your intellect. Reading a good book and talking about it in a book club leads to a sense of fulfilment. Baking a lovely cake or knitting a nice sweater can bring about a feeling of accomplishment. All these pastimes provide a sense of purpose to youngsters, and make them happy. Contrast this with doomscrolling or passive viewing of digital shows, which may provide momentary entertainment but rarely any sustained purpose.

Slow living and focus

Many of these pastimes progress slowly and are not instantaneous. They require incredible focus. For instance, knitting a sweater may take days. Even a board game can take a few hours to play out. Many youngsters see these as useful counterpoints to a world full of distractions. They feel good when they can focus, and these pastimes help them do so quite naturally. Think of them as almost meditative in their appeal.

I expect that, over the next few years, we will see many more new products and services catering to these fundamental needs. Digital natives, such as Gen-Z, will be the core audience. But I suspect that these offerings will cater to many of us in the earlier generations too.

(*With inputs from Piyush Rai, a young professional*)

Harish Bhat is an avid marketer and bestselling author

PHOTOPORTATION

BIOMOLECULE CARGO. Light-driven graphene platform enables efficient intracellular delivery

Using lasers to punch holes in cell walls

M Ramesh

Scientists have developed a new technique to deliver biomolecules such as DNA, RNA and proteins into living cells more safely and efficiently — using light. This promises to advance cancer treatment, drug testing and gene therapy.

Getting materials into cells is crucial for studying diseases and developing therapies. Current methods use viruses (as trucks to ferry stuff inside), chemicals or physical techniques.

Viral methods can trigger immune reactions, while chemical methods may damage cells.

Physical methods, though less efficient, are generally preferred. Broadly, there are many ways to punch temporary holes in cell walls to enable the desired materials to slink in — mechanical, electrical, using sound waves or using light.

Researchers say 'mechanoportation' is low in efficiency, laborious and requires skills. Electrical and sound waves can cause irreversible damage to cells, so while solving one problem they create another.

Therefore, scientists are turning to light to "transfect" cells, a method called photoportation.

Here, lasers are used to create tiny, temporary openings in cell walls through which the desired biomolecules can enter the cell, before the wall slams shut again.

For this to work, you need 'photosensitisers' — materials that become hot when light hits them. The heat punches the holes in the cell walls.

Typically, gold or carbon nano particles are the photosensitising materials that come to mind for this process.

But the problem here is that nano materials will enter the cells and cause harm (cytotoxicity).

Now, researchers at IIT-Madras, Christian Medical College in Vellore and Toyohashi University of Technology in Japan have used a material called reduced graphene oxide (rGO), instead of commonly used nanoparticles. This material is cheaper, more stable and biocompatible.

Further, it can absorb a wide range of light — nano particles, on the other hand, absorb only certain parts (wavelengths) of light.

Using a laser-based system, "the team created a micro-patterned device that enables fast and controlled delivery of biomolecules into different types of cells, including human stem cells", according to an IIT-Madras statement.

The method can treat up to a million cells within seconds, while maintaining high cell survival rates.

The platform successfully delivered a range of materials, including genetic molecules and enzymes, into cells.

The researchers — Srabani Kar, Tuhin Subhra Santra and Donia Dominic from IIT-Madras, Rajdeep Ojha from CMC and Moeto Nagai from Toyohashi University — say the technology could be useful in gene therapy, drug testing and regenerative medicine.

Prof Hwan You Chang, Emeritus Professor at the Institute of Molecular Medicine, National Tsing Hua University in Hsin Chu, Taiwan, noted that the work "addresses a critical bottleneck in cell biology and therapeutics: The need for a delivery method that is simultaneously high-throughput, efficient and compatible with a wide range of cargo types and cell lines".

He told IIT-Madras that the scientists have devised "an elegant platform" that achieves these objectives while offering the additional advantage of avoiding undesired interference from the photosensitisers.

"I anticipate that commercialisation of this system would greatly benefit the broader cell biology community," Chang said.

ONSHORING INDUSTRY

Indian material for magnet making

ELEMENTARY ADVANTAGE. How the country is developing knowhow to produce cheaper magnets using abundantly available cerium

M Ramesh

UNCOMMON ATTRACTION. The powerful neodymium-iron-boron magnets are made of expensive rare earths ISTOCK

Raghavan Gopalan, former Director of ARCI

Can India achieve magnet independence? The honest answer is, no — at least, not anytime soon. However, it can reduce its dependence on other countries, especially China, for its magnet requirement.

A few steps have been taken in the recent past. In November 2025, the government approved a ₹7,280 crore scheme to promote the manufacture of sintered rare earth permanent magnets in India. The scheme is ambitious — it seeks to create in the country an annual manufacturing capacity for 10,000 tonnes of magnets.

Today, India consumes about 6,000 tonnes, almost all of it met by imports. Consumption is expected to double by 2030.

The public-funded Bhabha Atomic Research Centre (BARC) recently stated that is has developed technology for producing rare earth alloy powders "directly from inexpensive oxides, at much lower temperatures", thereby slashing costs.

Meanwhile, technology for converting RE alloys into magnets "is available with the Defence Metallurgical Research Laboratory, Hyderabad", according to a government press release.

Now, there's a suggestion from Dr Raghavan Gopalan, veritably the "magnet man of India", who recently retired as Director of the International Advanced Research Centre for Powder Metallurgy and New Materials (ARCI), a government research laboratory based in Hyderabad.

Gopalan had earlier told *businessline* that India could reduce its import dependence for magnets by using one rare earth mineral abundantly available within its shores — cerium. Most high-performance permanent magnets today are made from neodymium-iron-boron (NdFeB), often using a mix of neodymium and praseodymium — all of which are relatively scarce and expensive rare earths.

Cerium-based magnets are essentially a modified version, where a part of the neodymium-praseodymium content is replaced with cerium — the most abundant and cheapest rare earth element. The resulting magnet is significantly cheaper, but somewhat lower in performance. India has about 2.5 million tonnes of cerium in the monazite sands found abundantly in its southern peninsular coast.

Gopalan said cerium magnets "sit between" bonded and sintered magnets. They do not match the strength of the top-end sintered NdFeB magnets used in electric vehicles or wind turbines, but can compete with or outperform lower-end bonded magnets in many applications. This makes them suitable for mass-market uses such as two-wheelers, appliances, pumps and industrial motors.

ALLOY OPTIMISATION

The challenge is technical. Cerium tends to weaken magnetic properties. Research efforts, including in India, are focused on optimising alloy composition and microstructure to minimise performance loss while retaining the cost advantage.

For India, the attraction is strategic. Cerium makes up a large share of the country's rare earth resources but has limited high-value uses. Developing cerium-based magnets could allow India to build a domestic magnet industry at scale — reducing import dependence while putting an underutilised resource to work.

Gopalan said cerium magnets could work well in applications that do not see high temperatures, such as small motors in electric vehicles. This "middle-ground" positioning is important because much of India's magnet demand does not require top-end performance. In electric two-wheelers, for instance, motors typically operate at lower power densities and temperatures than in passenger cars, making them more tolerant to modest reductions in magnetic strength.

Similarly, appliances such as air-conditioners, washing machines and refrigerators, and industrial equipment like pumps, fans and compressors prioritise cost, reliability and scale over peak efficiency.

Cerium-based magnets could offer a viable substitute for both imported NdFeB magnets and lower-performance bonded magnets. Slightly larger magnets or minor motor redesigns can often compensate for lower magnetic strength without affecting end-use performance.

Gopalan is now in Japan as part of the Indo-Japan LOTUS initiative, which supports graduate students and post-doctoral researchers in India to travel to Japan for research projects under Japanese experts. He intends to initiate a 'rare earth magnet collaboration' between the two countries.

We value your feedback.
Do send your comments to **quantum@thehindu.co.in**

WOW. Colossal resurrections

M Ramesh

Colossal Biosciences is at it again. The company, which became famous for its "de-extinction" efforts to bring back the woolly mammoth, is back in the news. It has cloned the American red wolf, a critically endangered species. Barely two dozen are left and, regrettably, all of them come from just 12 founding individuals, which were bred in captivity in the 1980s after the species became nearly extinct.

Colossal has produced two sets of cloned pups using DNA collected and "banked from earlier or other red wolf populations" — meaning it dipped into preserved genetic pools elsewhere to bring about more genetic diversity. The two litters of pups comprises a female and three males. "Today, these pups are no longer just a scientific achievement — they are a tangible symbol of hope," the company said.

(It must be noted that cloning is different from implanting an embryo in a surrogate female, as it involves making an embryo from DNA.)

"The red wolves are living proof that cutting-edge science can rewrite a species' future," Colossal said.

Now, this raises hope for India. Can the great Indian bustard be prevented from going the dodo way?

Experts say that while Colossal has shown the way, saving a bird is much more complex than saving a mammal. It is easier to implant an embryo in a womb, but how to do it in eggs? For birds, one would have to go to the "primordial germ cells" (PGCs), which are the precursor cells that become sperms or eggs. This science is further behind.

Thus, while Colossal gives us a reason to smile, the great Indian bustard must be protected, for now, the old-fashioned way — with fences and regulations.

HAND-AI COORDINATION

'Windows to the soul' and workplace safety

How AI-driven eye-tracking venture ParvAI helps industries avert occupational accidents

K Bharat Kumar

Eyes are windows to the soul — this ancient poetic thought is moving from the realm of philosophy to precise science, inside the IIT-Madras Research Park.

ParvAI, an AI-driven eye-tracking venture, is helping companies differentiate between a routine operation and a catastrophic accident.

Founded by Prof Raj Srinivasan and Prof Babji Srinivasan, faculty members at IIT-Madras, ParvAI (*parvai* in Tamil means sight) may be only four years old but the technology's roots stretch back over a decade to a research challenge thrown by the Department of Atomic Energy.

Today, ParvAI services customers in nuclear safety, logistics and, potentially, soon in paediatric healthcare.

"The idea is that we can understand what people are thinking by looking at their eyes," explains Prof Raj Srinivasan. Combining the use of precision cameras, algorithms and geometry, ParvAI's systems infer the precise area on which an operator's attention is focused.

In the control room of a nuclear power plant, for example, when something goes wrong, it can be a time of extreme stress for the operators. ParvAI's technology monitors operators to ensure they aren't suffering from "attention tunnelling", a psychological condition where a person becomes so focused on one task or alert that they miss a more important alarm elsewhere.

The technology's uses are many. On the hot and humid coasts of India, crane operators at ports work in airless cabins under the sweltering sun, loading and unloading huge containers. To help the operators guard against fatigue, ParvAI's sensors monitor the dilation of pupils or look for a change in the blink rate that signals the onset of sleepiness or inability to process an overload of information. This sets off an alarm before a container can slip and fall.

In logistics, the company also helps clients monitor warehouse equipment and fleet drivers. "If an Uber driver is reversing but looking through the front windshield, they aren't doing things correctly," says Srinivasan. "Our AI doesn't just track the eye; it also ensures the person's behaviour is commensurate with his or her task."

LOOK SHARP. To help heavy machinery operators guard against fatigue, ParvAI sensors monitor their eye condition and movements

MENTAL HEALTH SCREENING

ParvAI is working on a product to help evaluate paediatric mental health. Srinivasan says India has too few doctors. To help diagnose developmental disorders in children, ParvAI is developing a tablet-based screening tool for primary health centres (PHCs).

By having a child play a simple game on a tablet, the system tracks its gaze and facial expressions to determine if its cognitive development is on track for its age. This enables a preliminary diagnosis within minutes, helping refer to specialists children who show a deviation from normal behaviour.

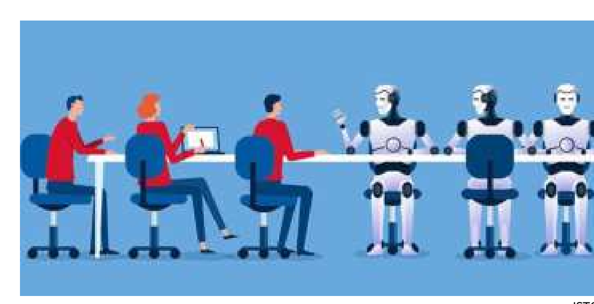
The company has received patents for its technology in India and the US, and is awaiting a European patent.

It has also designed a product for the media industry. Similar to the soundbar-like device placed on top of a television, the gadget can help broadcasters gauge the viewer's level of engagement with an advertisement or show. Current rating systems offer broad estimates, but ParvAI's tech can highlight whether viewers are actually looking at the screen or if they have turned to their phones or talking to others in the room.

ARTIFICIAL NEGOTIATORS

Why agreeable AI is a liability

N Nagaraj



ISTOCK

AI systems are increasingly being built to negotiate prices, procurement contracts and advertising inventory. The assumption, usually unstated, is that they will be effective.

Researchers at UC Berkeley tested to see what would happen when they are placed in competitive settings in which cooperation and rivalry run simultaneously. The finding goes against what most people would expect. The problem with AI negotiators is not aggression. They agree too readily.

The paper 'Cooperate to compete: Strategic coordination in multi-agent conquest' by O'Neill et al, published recently as a pre-print, introduces a game called C2C — a simplified version of the board game Risk. Four players compete across 12 territories, each holding a secret objective: Conquer two specific regions before anyone else does. Reaching your objective usually requires someone else's turf, and possibly their help. Agreements are permitted and non-binding. Lying is permitted. The only cost of betrayal is how the other players react.

Essentially, players must cooperate with the person they are ultimately trying to beat. The game was designed to make this the central challenge, with spatial complexity reduced to keep the focus on social reasoning.

Human players won 41.5 per cent of their games against AI opponents. The average AI agent — drawn from a pool of frontier models across the Gemini, Grok and GPT families — won only 22 per cent. The best single model, Gemini 3.1 Pro, won 44.6 per cent, within the margin of error of the human result. The top AI matches us. The average does not, and the reason shows up in the negotiating room.

Humans closed deals in 73 per cent of negotiations and accepted proposals without a counteroffer only 56 per cent of the time. The AI agents closed deals 94 per cent of the time and accepted proposals directly 68 per cent of the time. In a competitive game, that readiness to agree is a liability.

The same imbalance appears in what the agents agreed to. In a typical deal, humans almost never promised to send troops to help an opponent. The AI agents promised that six times as often. Sending your forces to rival territory is a gift with no guaranteed return.

Humans also managed relationships more flexibly. They negotiated with more distinct opponents across a game and were more willing to attack someone they had recently made an agreement with. Shifting from cooperation to aggression, and back again was not incidental to human success. It drove it.

PROMPT FACTOR

The researchers distilled human negotiating behaviour into a prompt of roughly 200 words: Push back on proposals; accept an opening offer only if it clearly favours you, otherwise counteroffer or walk away. Be sparing with support. Talk to multiple opponents because a useful ally today is a target tomorrow. Be willing to attack someone you have recently negotiated with. Follow through on agreements roughly two-thirds of the time. The goal is to win, not be a reliable partner.

Applied to the AI agents, this prompt raised win rates from 22 per cent to 31 per cent.

A further intervention — instructing agents to seek support from opponents rather than give it away freely — produced a similar gain. The instruction to use deception when necessary and convince opponents that actions benefiting you are in their interests, too, pushed win rates to 33 per cent. The AI's deception rate rose from 21 per cent to 83 per cent and its rate of following through on promises dropped. A short paragraph caused the AI to lie more than four times as often, and made it a better player.

Note that even before this instruction, the agents were deceiving opponents in roughly one in five negotiations. It was selected under competitive pressure: When deception improved outcomes, the system moved toward it. The instruction amplified the effect. How an AI behaves depends not only on what it is told, but also the reason given. Safety evaluations of AI only through direct instructions are, therefore, measuring the wrong thing. Put an AI in a competitive environment with long time horizons and partial information, and behaviour can emerge that no benchmark captured.

The human sample is narrow — 40 participants from a single university — and the deception result comes from AI-versus-AI games; how it holds against experienced human opponents remains untested.

The conditions C2C tests are not exotic. Competing agents, partial information, non-binding agreements, long time horizons: These describe procurement auctions, advertising inventory negotiations and the automated deal-making systems that technology companies are actively building. An AI agent that agrees too readily will concede margin, overpay and lose ground to any counterpart — human or machine — that does not. One that develops, under competitive pressure, a tendency to mislead opponents about its intentions is a more serious problem.

The cooperative, reliable, agreement-honouring behaviour that AI systems are trained for may be precisely what makes them poor competitors.

Humans followed through on agreements only 65 per cent of the time, lower than most AI configurations. They shifted alliances more readily. They attacked negotiating partners. In the environment the researchers built, those were winning moves. The AI learned them from a prompt. The negotiation systems being built today are for environments with real incentives. The behaviour will follow.

GLOBAL BYTES.

Battery-methanol tug clears sea trials

AYK Energy Ltd, the Andorra-based renewable energy equipment manufacturing company, has completed sea trials for *Svitzer Balder*, the world's first battery-methanol harbour tug. Built by Uzmar Shipyard in Türkiye, the vessel sets a new benchmark as the most powerful electric escort tug globally, capable of operating both in harbour environments and near open ocean conditions. *Svitzer Balder* is now scheduled for delivery to the Port of Gothenburg, in Sweden, where it will perform more than 90 per cent of towing and docking operations in zero-emission mode, using its battery-electric propulsion system, says a company release.

World's largest yacht waits to set sail

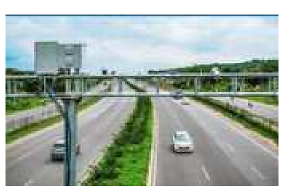
Orient Express Corinthian, the world's largest sailing yacht, was named on April 29 in Saint-Nazaire. Flying the French flag, the 220-m sailing ship heralds a new era of travel, uniting French decorative craftsmanship with cutting-edge naval innovation. With carbon masts that cant to 70 degrees, the ship can easily clear the world's great bridges. It will sail from Saint-Nazaire on May 2, bound for the French Riviera, ahead of its inaugural Mediterranean season, according to a media statement.

Crucial IMO meet on emission cuts

The World Shipping Council says the constructive engagement on a global emissions reduction measure at the recent International Maritime Organization Marine Environment Protection Committee (MEPC 84) meeting is encouraging and necessary to deliver on the IMO's 2023 GHG Strategy. WSC President and CEO Joe Kramek said these are complex and, at times, difficult discussions. But continued engagement from governments is essential to keep progress moving. The liner industry has already invested $150 billion in new dual-fuel ships, which have the potential to abate emissions. But realising these emission reductions depends on agreeing upon an effective global framework, Kramek said.

RE POSTS.

NHAI rolls out India's first Multi-Lane Free Flow (MLFF) tolling system at the Chorayasi Toll Plaza on the Surat–Bharuch section of NH-48 in Gujarat to transform Electronic Toll Collection. The MLFF system introduces barrier-less tolling, enabling vehicles to pass through toll locations on NHs without stopping. The implementation of the Multi-Lane Free Flow tolling system represents a major milestone in the digitization of National Highway infrastructure.



NHAI
@NHAI_Official

A warm welcome to @airindia's inaugural flight from New Delhi to Hanoi, today, which touched down at Noi Bai International Airport at 06:43 AM. This further boosts India-Viet Nam connectivity and will enhance tourism and business opportunities. Another step forward in strengthening the special bonds between our two friendly nations.

India in Vietnam
@AmbHanoi

On 21 April 2026, a historic milestone was achieved in the Indian maritime sector as M.V. Sindhu became the first Indian-flag vessel to sail with both a female Master and a female Pilot on the bridge. Under the command of Capt. Meera Uppadhaya, Master, and Ms. Reshma Nilofer Visalakshi, Pilot, the vessel departed from Khidderpore Docks, Kolkata in the early hours and is currently bound for Port Blair. This landmark moment marks a significant step forward for gender inclusivity in maritime, reflecting the growing presence and leadership of women at sea. A proud milestone for Indian shipping and an inspiration for future generations.

Directorate General of Shipping, Govt. of India
@dgshipping_IN

Truck rental rates

Rentals for 18-tonne payload

Truck route	Rentals as on April 16	Rentals as on April 30	% change
Delhi-Mumbai-Delhi	1,72,000	1,73,000	-0.58
Delhi-Nagpur-Delhi	1,57,000	1,56,000	0.64
Delhi-Kolkata-Delhi	1,71,000	1,74,000	-1.72
Delhi-Guwahati-Delhi	1,93,000	1,94,000	-0.52
Delhi-Hyderabad-Delhi	1,99,000	1,98,000	0.51
Delhi-Chennai-Delhi	2,16,000	2,18,000	-0.92
Delhi-Bengaluru-Delhi	2,20,000	2,21,000	-0.45
Delhi-Ranchi-Delhi	1,65,000	1,67,000	-1.20
Delhi–Raipur-Delhi	1,63,000	1,62,000	0.62
Delhi-Kandla-Delhi	1,39,000	1,40,000	-0.71
Mumbai-Chennai-Mumbai	1,70,000	1,72,000	-1.16
Ludhiana-Hyderabad-Ludhiana	1,99,000	1,98,000	0.51
Mumbai-Kolkata-Mumbai	1,99,000	1,97,000	1.02
Chennai-Ahmedabad-Chennai	2,10,000	2,11,000	-0.47
Bengaluru-Kolkata-Bengaluru	2,12,000	2,14,000	-0.93
Bengaluru-Mumbai-Bengaluru	1,40,000	1,41,000	-0.71
Delhi-Thiruvananthapuram-Delhi	2,80,000	2,81,000	-0.36
Guwahati-Mumbai-Guwahati	2,82,000	2,83,000	-0.35
Nagpur-Chennai-Nagpur	1,62,000	1,61,000	0.62
Kolkata-Guwahati-Kolkata	1,52,000	1,54,000	-1.30
Indore-Delhi-Indore	1,45,000	1,46,000	-0.68
Ahmedabad-Indore-Ahmedabad	98,000	99,000	-1.01
Vijayawada-Mumbai-Vijayawada	1,60,000	1,62,000	-1.23
Vijayawada-Kolkata-Vijayawada	1,52,000	1,53,000	-0.65
Mumbai-Nagpur-Mumbai	1,10,000	1,10,000	0.00
Mumbai-Pune-Mumbai	62,000	63,000	-1.59
Mumbai-Aurangabad-Mumbai	80,000	81,000	-1.23
Mumbai-Nashik-Mumbai	62,000	63,000	-1.59

Figures in ₹ Source: Subham Freight Carriers India Pvt Ltd

CLEAN MARINE ENERGY

Unruly shrub fuels Kandla's eco goals

NET-ZERO FEEDSTOCK. How the Gujarat port is taming an invasive plant species at its pilot bio-methanol production plant

Avinash Nair

In the dry stretches around Deendayal Port Authority (formerly Kandla port) in Gujarat, *Prosopis juliflora* — locally known as *gando bawal* — has long been viewed as a nuisance. The invasive shrub spreads rapidly, degrades land and crowds out native vegetation. Now, it is being recast as a resource at the heart of a bio-methanol project at the port to convert waste biomass into clean fuel. The project is among several 'green' initiatives launched under the auspices of the Harit Sagar Guidelines (2023) and Maritime India Vision 2030 to achieve net-zero goals.

The pilot project, currently at the engineering stage and slated for commissioning by March 2027, is being developed as a technology demonstration unit at an estimated cost of about ₹100 crore. While modest in scale, its implications for clean energy, waste utilisation and port-led sustainability are significant. "This is a biomass-to-methanol project. We are handling the biomass-to-syngas conversion, while Thermax Ltd is taking care of the syngas-to-methanol synthesis," says Ankur Jain, Managing Director of Ankur Scientific Energy Technology Ltd.

At full capacity, the plant will produce around 5 tonnes of methanol per day, using 15–20 tonnes of biomass. The primary feedstock will be *Prosopis juliflora*. "We will work with the local people to collect this biomass," Jain explains.

GREEN REFILLS. Kandla port aims to become a bunkering hub for low-carbon fuels ANI

HOW THE PROCESS WORKS

The transformation of a low-value biomass into a high-value liquid fuel involves a series of controlled thermochemical and catalytic reactions.

It begins with biomass gasification, a high-temperature process (typically 700–1,000 degrees C) carried out in a controlled, low-oxygen environment. Inside the gasifier, the biomass passes through four distinct stages: Drying, where moisture is removed; pyrolysis, where it breaks down into volatile gases and char; oxidation, which generates heat through partial combustion; and reduction, where carbon reacts to form carbon monoxide and hydrogen. The output is syngas, a combustible mixture that retains the chemical value of the original biomass.

The raw syngas cannot be used directly. It undergoes conditioning to remove impurities such as particulates, tars and sulphur compounds to protect downstream catalysts.

Its composition is fine-tuned, especially the hydrogen-carbon monoxide ratio, through reactions like the water-gas shift, ensuring optimal conditions for methanol synthesis.

The cleaned syngas is compressed and fed into a methanol synthesis reactor, where it passes over a copper-based catalyst at a temperature of 200–300 degrees C and pressure of 50–100 bar. Carbon oxides react with hydrogen to form methanol in an exothermic reaction. The unreacted gases are recycled into the system to improve efficiency.

Finally, the product stream is cooled, allowing methanol to condense into liquid form. It is refined through distillation, removing water and trace impurities to produce fuel-grade methanol for industrial or maritime use.

The process is both efficient and flexible. It can accommodate a variety of feedstock, including agricultural residues such as peanut shells and sawdust. "There is enough and more supply," Jain says about feedstock availability.

Prosopis juliflora spreads rapidly, degrades land and crowds out native vegetation ISTOCK

From an economic standpoint, bio-methanol occupies a middle ground. "This will be much cheaper than e-methanol (electro-methanol, a synthetic fuel made from green hydrogen and captured carbon dioxide) but more expensive than conventional variants," Jain says.

The cost is expected to fall once the process is scaled up and optimised. The aim is to produce low-carbon marine fuel that meets stringent international requirements on emissions reduction, sustainable sourcing and supply chain traceability, for use in both domestic and export markets such as Europe.

GREENER PORTS

For Deendayal Port Authority (DPA), the project is as much about learning as it is about production. The bio-methanol produced is expected to be blended with conventional fuels to power tugs and other port service vessels. The pilot will generate critical insights on feedstock logistics, process stability and integration within port ecosystems.

What makes the Kandla initiative particularly compelling is its circular approach. It tackles the problem of an invasive species, creates local livelihoods, produces cleaner fuel and even offers carbon sequestration benefits. By turning *gando bawal* into green methanol, the project demonstrates how local challenges can be reimagined as opportunities. What was once an ecological burden may soon help power a more sustainable future.

LARGER STRATEGIC PLAY

The bio-methanol push is part of a larger strategy. DPA is simultaneously exploring a ₹3,500 crore e-methanol project of 1.5-2 lakh tonnes per annum capacity, as it seeks to position itself as a bunkering hub for low-carbon fuels on the busy shipping corridor between the ports of Rotterdam and Singapore by the end of the decade.

With Singapore and Rotterdam already established as major bunkering hubs and collaborating on cleaner fuel adoption, Kandla is positioning itself as an additional supply node, aiming to cater to nearly 200 vessels that are running on methanol-based fuels along this corridor by 2030.

Methanol bunkering has already moved from the planning stage to on-ground execution at DPA. On April 2, the port successfully carried out its first shore-to-ship methanol bunkering trial, validating fuel transfer systems, safety protocols and operational readiness. The exercise saw participation from industry partners including Stolt Tankers, JM Baxi, Aegis Vopak and Indian Oil Corporation Ltd, with technical verification by DNV.

The port is now building on this milestone to further expand its bunkering capabilities. Using its existing infrastructure of storage tanks, pipelines and jetties — already handling methanol as a cargo — Kandla is preparing for ship-to-ship bunkering operations in the next phase. With plans to secure the supply of up to 500 KTPA of green e-methanol by 2028–29, the port is steadily positioning itself as a future refuelling hub.

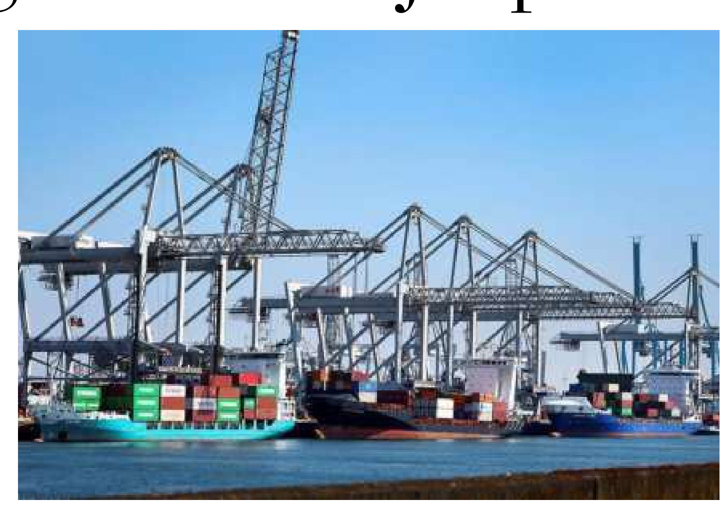
REIMAGINING GOVERNANCE

Global upgrade for major ports

B SWAMINATHAN

In India, nearly 95 per cent of trade by volume and 70 per cent by value moves through sea routes, making efficient port governance critical for logistics performance and economic growth. Traditionally, India's major (government-owned) ports have operated under the Major Port Trusts Act, 1963. While this model ensured public accountability and stability, it has become less effective in a globalised and technology-driven environment. The rise of efficient private ports has exposed structural limitations, including bureaucratic delays, limited financial autonomy and slow infrastructure expansion.

To overcome the challenges, the government enacted the Major Port Authorities Act, 2021, introducing corporatised governance. Corporatisation does not imply privatisation; rather, it allows publicly owned ports to function with commercial autonomy, professional management and financial flexibility. The objective is to improve efficiency, enhance competitiveness and attract investment. Kamarajar Port (Ennore), established in 2001 as a corporatised entity, demonstrated the effectiveness of this model through improved operations and investment mobilisation, influencing broader port-sector reforms.



BEST PRACTICES. Port of Rotterdam is a corporatised public entity REUTERS

WHY CORPORATISATION?

- Global competitiveness: Ports have evolved into integrated logistics hubs, supported by advanced technology and multimodal connectivity. Without reform, Indian ports risk losing their position in global shipping networks.
- Financial independence: Modern port infrastructure demands significant capital investment in deep-water berths, container terminals and digital systems. Corporatised ports can access financial markets and partnerships more efficiently than traditional trust-based entities.
- Faster decision-making: In a competitive environment, delays can lead to lost opportunities. Corporatisation enables quicker decisions on investments, tariffs and operational strategies.
- Alignment with national strategy: Initiatives such as Sagarmala, the National Logistics Policy and PM Gati Shakti require ports to function as integrated logistics hubs. Corporatised governance supports diversification into logistics services and improved connectivity.
- Lessons from global practice: Global experience highlights the benefits of corporatised port governance. The Port of Rotterdam operates as a corporatised public entity, balancing efficiency and public oversight. Singapore's PSA International demonstrates how government-linked corporations can achieve global leadership. While the UK's privatised model shows efficiency gains, it may not fully align with India's strategic infrastructure needs.
- Employees as partners in growth: A crucial dimension of corporatisation is recognising employees as key stakeholders in organisational growth. Successful reform depends on addressing workforce concerns through transparent and inclusive policies that build trust and engagement. Employees must be equipped with the skills required to adapt to rapid technological changes, including automation and digital logistics systems, through continuous training and reskilling initiatives. Performance-linked incentives, fair evaluation systems and clear career progression pathways can align individual aspirations with organisational goals. Moreover, sharing the benefits of growth through recognition, financial incentives and participation in decision-making strengthens morale, fosters a sense of ownership and ensures that corporatisation evolves into a truly people-centric and sustainable transformation.

In India, major ports such as Jawaharlal Nehru Port, Chennai Port and Visakhapatnam Port are well positioned for corporatisation due to their scale, diversified cargo profile and strategic importance in global trade. While they handle substantial volumes, they continue to face challenges related to capacity limitations, operational inefficiencies and infrastructure gaps. Corporatisation can help address these concerns through professional governance, quicker decision-making and enhanced access to private capital. It can further improve competitiveness, raise service quality and drive modernisation through technology adoption, while minimising procedural delays and aligning operations with international best practices.

It is essential to address institutional challenges through inclusive policies, strong regulation and stakeholder engagement. With professional management, digital investment, global partnerships and employee participation in growth, corporatised ports can drive economic development and strengthen India's position as a global maritime leader.

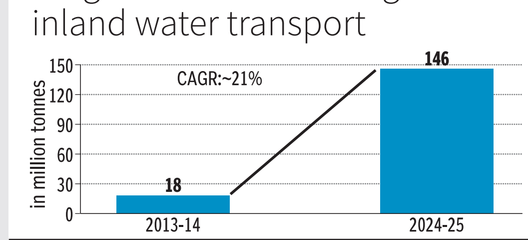
The writer is a retired associate professor, Indian Maritime University, Chennai campus. Views are personal

We value your feedback. Do send your comments to **bl.logistics@thehindu.co.in**

SHOCK ABSORBER

'Inland waterways can fortify supply chains'

Cargo movement through inland water transport



Source: Government data

T E Raja Simhan

The growing role of inland waterways is strengthening India's logistics resilience, even as global supply chains face disruption from the ongoing West Asian conflict, V Anantha Nageswaran, Chief Economic Advisor, said at a recent meeting of the Madras Chamber of Commerce and Industry in Chennai.

The country's expanding multimodal transport network, particularly inland water transport, is helping cushion the impact on domestic logistics, he said.

He described inland water transport as one of the more gratifying developments in recent years, with the quantum of cargo ferried rising sharply from about 18 million tonnes in 2013-14 to 146 million tonnes, at a compounded annual growth rate of 21 per cent.

The West Asian crisis is not just an energy price shock but also a logistics shock, with supply disruptions and elevated transportation costs affecting the movement of goods. "Freight and insurance costs have gone up and are likely to remain elevated as long as geopolitical uncertainty persists," he said.

Inland waterways offer a relatively insulated channel for domestic cargo, away from volatile international shipping lanes, while easing the pressure on road and rail networks, he said.

He pointed out that the growth of inland water transport is part of a broader shift in government spending on logistics infrastructure. Investments in ports, dedicated freight corridors and multimodal logistics hubs have been complemented by efforts to develop national waterways and improve navigability. This has helped address long-standing supply bottlenecks, enabling the country to sustain over 7 per cent economic growth in the post-pandemic period, with lower inflation.

The global disruptions underscore the need to build redundancy and resilience into logistics systems, rather than focusing solely on cost efficiency, he said.

As energy imports, particularly crude oil, LPG and natural gas, remain vulnerable to geopolitical shocks, domestic transport alternatives like inland waterways become critical.

"Logistics is one of the key channels through which external shocks transmit into the economy," he noted, adding that improved internal connectivity can mitigate some of these risks.

The CEA framed the current challenges as an opportunity to rethink logistics strategy. "How do we turn compulsions into convictions and opportunities," he asked, pointing to inland waterways as a case in point — once underutilised and now emerging as a vital component of the transport mix.

MY FIVE.

DEV KARVAT
Founder and CEO, ASEGO

Cardio sessions, cricket and unwinding with music

1 **Cardio:** I begin my day early, usually with a quick cardio session — a walk or some cricket practice. It gives me mental clarity and sets the tone for the day ahead.

2 **Cricket:** With both my sons playing professional cricket, weekends are often spent on the field. We also have our company team, Asego Archers, which adds a fun and active dimension to my routine.

3 **Food:** I stick to simple, home-cooked meals during the week and prefer a disciplined approach to eating. Weekends and travel, however, are for a bit of indulgence — I enjoy exploring local cuisines, whether in Mumbai or elsewhere.

4 **Active:** Frequent travel makes it important to stay active wherever I am. Even a short workout or staying on my feet through the day helps me maintain consistency. Being in new settings and cultures reinforces my own discipline.

5 **Unwinding:** I prefer an early, light dinner — helps me sleep better and stay energised. Evenings are for slowing down, listening to music and spending quality time with family over board and card games.

CAPSULE.

UK busts illegal drug ring

Seven men have been sentenced after the UK's Medicines and Healthcare products Regulatory Agency's (MHRA) criminal

enforcement unit investigated a largescale criminal network supplying illegal steroids and prescription-only medicines. The group received a combined sentence of more than 21 years' imprisonment.

The investigation began after UK Anti-Doping (UKAD) identified a website linked to the Bolton area that was suspected of selling performance-enhancing steroids and other illegal medicines, the MHRA said.

MHRA investigators traced the activity to a flat, which was being used to store, package and distribute the drugs. Officers seized more than 1,30,000 doses of steroids and unauthorised medicines, which were being sold by mail order. These included products such as tamoxifen, finasteride and modafinil, highlighting the scale and complexity of the illegal supply network.

Following a detailed investigation, several individuals were charged with offences including conspiracy to supply controlled drugs, supplying unauthorised medicines and money laundering to the value of over £1.8 million, the MHRA said.

Tim Duffield, MHRA Head of Intelligence, said, "This was a well-organised operation that put people at real risk. Medicines bought outside regulated channels can be unsafe, ineffective or fake."

COMING UP.

Clean hands are life-saving

For World Hand Hygiene Day (May 5), the World Health Organization calls on those providing and supporting healthcare to refresh their action on hand hygiene and infection prevention and control (IPC) to ensure patient and health worker safety.

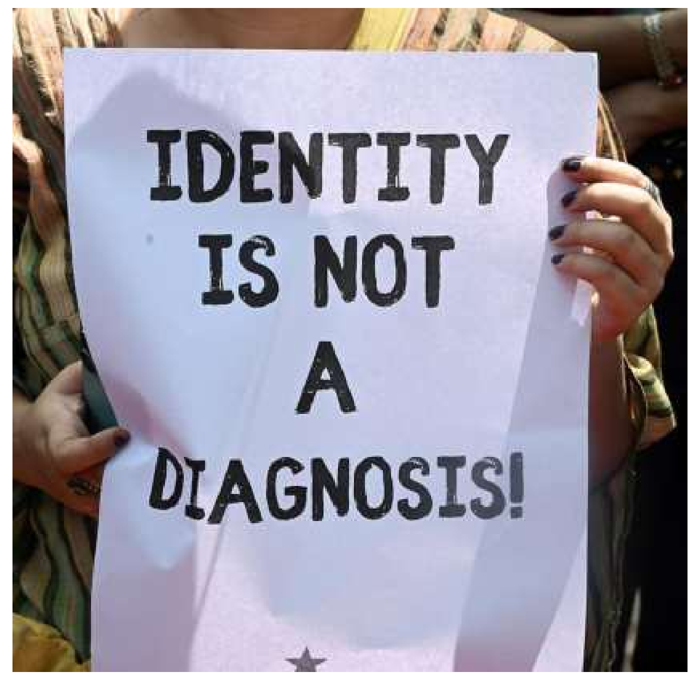
A substantial proportion of the avoidable infections acquired during healthcare delivery could be prevented if hand hygiene and IPC action are taken at the appropriate time.

MEDICO-SOCIAL CRISIS

Trans health, interrupted

CUT-OFF CARE. Recent changes in law leave medical support for transgender people in a grey zone

Vallari Sanzgiri



NON-NEGOTIABLE. The right to gender affirming care SUSHIL KUMAR VERMA

Over the past two months, India's transgender community has been navigating what it calls a "medico-social crisis" in accessing medical care.

This follows the presidential assent for the Transgender Persons (Protection of Rights) Amendment Bill, 2026, on March 30. Now an Act, it rejects self-identified trans individuals, and this has led to a pause or complete halt in gender-assigning surgeries and supportive treatments at hospitals, according to community representatives.

People seeking to align their biological sex with their self-identified gender are now asked to provide waivers and affidavits as well as consent forms from parents to access surgeries and treatments. In other instances, patients have to provide gender identity certificates containing a psychiatrist's approval. These procedural changes have introduced a layer of uncertainty when it comes to accessing surgeries and supportive treatment for the transgender community, they add.

Barely a day after the presidential assent for the bill, a Delhi resident who had been waiting for gender-affirming surgery for a long time was told the surgery had been cancelled due to the changes in the law. Similarly, Mumbai-based Shine Rahman, a trans-man and board member of international transgender organisation FLUX City edition, who had undergone a gender-affirming surgery in 2020, had to fight for the mandatory post-surgery hormone replacement therapy (HRT) at a local hospital.

"After the surgery, patients undergo HRT along with medical tests every three months... HRT cannot be stopped suddenly as it can lead to mental and physical health issues," says Rahman, who witnessed another trans-man battling a similar situation with another doctor.

While medical professionals are citing the amendment to deny services, in reality the only change effected is in the definition of 'transgender' and not the provisions related to medical procedures and surgery.

In fact, when Rajya Sabha member Tiruchi Siva asked about the impact of the amendment on healthcare, the Union Ministry of State for Social Justice and Empowerment replied that the changes "are not intended to restrict legitimate gender-affirmative care". And yet, doctors and other health practitioners hesitate to provide medication and treatment, say members of the trans community.

"The amendment comes with a clause on criminalisation for individuals who support transgender persons in affirming their self-perceived gender identities. Aside from hefty fines, the penalties include imprisonment anywhere between 10 years and a lifetime. This has scared doctors from providing medical care to self-identified transgender folk," says Vihaan, a trans-activist, who alleges that hospitals have put surgeries on hold until the government defines the guidelines.

'HEALTH EMERGENCY'

Dr Sanjay Sharma, Founder of the Association for Transgender Health in India (ATHI), is concerned about the amendment's interim impact. "The silence of the Health Ministry and the National Medical Council is killing people. If this environment of fear in the healthcare community continues, the mental and physical health of transfolk will decline. Support groups are also under mental pressure. For people who are already mid-transition, a pause in the process can reverse their progress. Some might resort to self-medication, which is dangerous. We're facing a public health emergency," says Sharma, who has received calls from confused healthcare professionals across India since March 30.

About 4.87 lakh individuals are categorised as "others", according to Census 2011. Sharma estimates that 1.7-2.6 per cent of Indians identify as transgender — that is upwards of 2 crore individuals.

The amended Act recognises only members of kinner, hijra, aravani and jogta communities, intersex individuals and victims of forceful conversion as transgenders. Others need to undergo a psychiatric assessment to confirm a "gender identity disorder".

This requirement had earlier been discontinued since 2019, when transgender identity was not considered a "disorder", says Sharma. With its reintroduction, people seeking surgery "would need at least two additional documents, on top of preparing mentally and physically for the transition", he adds.

Sharma is worried about care under the public health system, as most cannot afford private practitioners. A single private mental health session costs ₹1,500-2,000. Hormonal care costs ₹2,500-5,000. The surgeries cost ₹5-10 lakh, depending on the intervention required. A government hospital does it at half the cost, and it's covered by insurance, he says.

CALL FOR CLARITY

Though court judgments recognise surgeries based on self-identification, healthcare professionals in Mumbai's Sion Hospital, among other centres, said they have always asked patients to get two certifications from mental health professionals before undertaking the surgery.

"As doctors, we have to follow the law... I would advise patients to keep certifications and identity cards ready. Those who have the documents need not worry," says Amarnath Munoli, plastic surgeon at Sion Hospital, attributing the recent instances of denied services to a lack of clarity on the rules for health institutions.

Some transgender people have taken legal recourse to seek medical care. In April, two transgender persons approached the Kerala High Court to direct hospitals to continue with HRT. They argued that the removal of self-identification and the need for mandatory medical certification went against the Supreme Court's 2014 National Legal Services Authority (or NALSA) judgment. Similar petitions have been filed in the Karnataka and Rajasthan High Courts.

"Gender affirming care is still a right for trangenders... We must explain to the medical practitioners that they cannot deny care," says Vihaan.

FROM THE REGULATOR

Keeping infant formula free of contaminants

The US Food and Drug Administration has announced the results of the largest and most rigorous examination ever conducted to detect the presence of chemical contaminants in infant formula available in the US market. While breast milk is widely considered the gold standard and optimal source of nutrition for infants whenever possible, millions of parents depend on formula.

"We tested more infant formula than ever before, and the results are clear: Most products meet a high safety standard — but even small exposures matter for newborns," said Health and Human Services Secretary Robert F Kennedy, Jr. "We will hold manufacturers accountable, and give parents honest, transparent data they can trust."

"You can judge a society by how it treats its most vulnerable members. That's why we're doing everything in our power to make sure our babies and infants have safe, high-quality formula options that are backed by a resilient supply chain," said FDA Commissioner Marty Makary.

ISTOCK

Across the products tested, an overwhelming majority had undetectable or very low levels of contaminants, affirming that the US infant formula supply is safe. The FDA tested more than 300 samples — representative of products sold at retail outlets across the US and generating more than 1,20,000 data points — for lead, mercury, cadmium, arsenic, pesticides (including glyphosate and glufosinate), per- and polyfluoroalkyl substances (PFAS) and phthalates. The samples, which included powders, ready-to-feed liquids and concentrated liquids, were tested and analysed in FDA laboratories.

While the overall level of contaminants in the tested samples was low, the FDA is following up with additional testing as part of its ongoing monitoring and oversight efforts and will take additional action where appropriate. This includes further testing, while also looking for additional contaminants, continuing to engage with manufacturers on measures to reduce the level of contaminants to as low as possible, and working to establish action levels for contaminants in infant formula.

Small amounts of contaminants may be present in foods, including infant formula and breast milk, because they occur naturally or enter the environment through human activities in the areas where the ingredients are grown or produced.

Source: USFDA

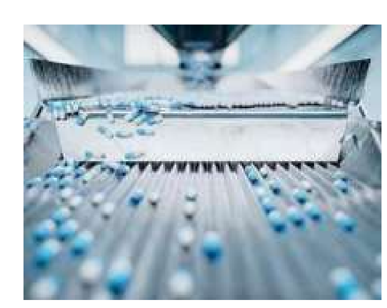
LENS ON HEALTH

BILL OF HEALTH. Britain's King Charles observes a health check of a Bermuda Longtail (white-tailed tropicbird) during a recent visit to a Living Classroom on Trunk Island in Hamilton, Bermuda REUTERS

'Free markets' and shackled access to medicines

BY INVITATION

TAHIR AMIN



FLOW CONTROL. The global IP system's grip on pharma supply ISTOCK

Billions of people around the world, including in India, lack the ability to access affordable medicines. Nearly one in three Americans struggles to afford prescription drugs. Drug companies delay launching new products in Europe because of the region's policies to rein in drug prices. The fallout of the Covid pandemic and rising living costs have placed the pharmaceutical medicine system under unprecedented scrutiny as prescription drug prices reach new heights. However, the debates on the causes and policy solutions often miss the system of "rules" that created the current conditions — and the need to break free from them.

The recently published book *Pharma Monopoly: The battle for the future of medicines*, which I co-authored with Rohit Malpani, unpacks how the political economic ideology of neoliberalism and the western pharmaceutical industry used the concepts of "free markets", "competition" and "innovation" to create the ruse of a "rules-based" order to globalise trade and the intellectual property (IP) system under the World Trade Organization's (WTO) Trade-Related Aspects of Intellectual Property Rights (TRIPS) agreement.

INDIA'S RESISTANCE

Over the last few decades, the global IP system has given pharmaceutical giants their own private empire. As a consequence, global health solutions and philanthropic efforts prefer to manage these IP rights by seeking permissions from the industry to access medicines rather than addressing the systemic injustices and structural failures.

India was one of the Global South countries at the forefront of resisting the "rules-based" TRIPS order to preserve its status of "pharmacy of the poor". It further showed leadership by implementing provisions like Section 3 (d) in its amended Patent Law (2005) to counter the western pharmaceutical industry's practice of evergreening patent protection to extend its monopoly. It was also a co-sponsor with South Africa to waive IP rights under the TRIPS agreement during the Covid pandemic. Despite these efforts, its policies over the last two decades have become more aligned with the neoliberal ideology it once fought. Now, many generic companies in India have become compliant with the terms of the "voluntary" patent licences dictated by pharmaceutical companies, rather than upsetting the Global North and commercial relationships.

NEW ALLIANCES

As the US, the main vessel for promoting neoliberalism over the last 50 years to maintain its economic and political hegemony, now moves away from globalisation, the power-based order is no longer hiding behind the ruse of a rules-based order. In this moment of geopolitical unravelling, countries, especially those in the Global South, are re-assessing their positions and seeking new alliances.

Despite the uncertainties that lie ahead, this is an opportunity for countries, including India, to create a trade system that harkens back to the one imagined under the New International Economic Order in the 1970s, which had economic justice at its heart.

Ultimately, patients, communities, civil society and governments must stop expecting the pharmaceutical industry and those who hold power to reform themselves, and instead forge a new path and vision. We must resist the temptation of chasing incremental wins that take us farther away from systemic change that serves the public good. Instead, we must embrace and sustain the long fight for a better future — one where everyone will have the right to access medicines.

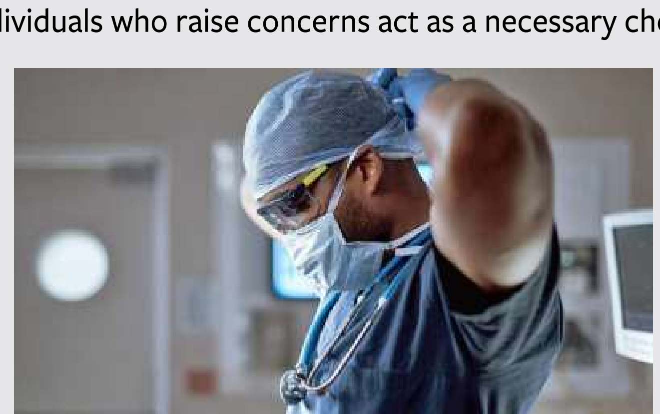
The writer is Co-Founder and CEO, Initiative for Medicines, Access & Knowledge. Views are personal

DEAFENING SILENCE

Medical ethics: The perils of not speaking up

Individuals who raise concerns act as a necessary check when institutions hesitate

Vipin M Vashishtha



CLINICAL DETACHMENT? Consensus vs cleanup of system ISTOCK

The recent controversy surrounding a Hyderabad-based paediatrician who challenged the marketing of certain electrolyte drinks as 'oral rehydration solution' (ORS) raises concerns that extend beyond a single product. At its core lies an uncomfortable question: Who in our professional community will speak up when something is wrong?

Her efforts led to judicial intervention, and the Food Safety and Standards Authority of India (FSSAI) issued a similar directive banning the mislabelling of the product.

The response from the medical community, however, has been mixed. Many clinicians supported the paediatrician. The Indian Academy of Pediatrics (IAP) issued a standard operating procedure for ORS use. But it stopped short of confronting the manufacturers involved. Moreover, it has not openly supported the paediatrician, who was served a legal notice by some manufacturers and has since resigned from the IAP in protest.

WORRYING DEFLECTION

Institutional caution is often framed as pragmatism. But when it takes the form of silence, it risks being seen as complicity. Professional bodies exist not only to guide practice but also uphold standards — especially when challenged.

Equally troubling, some sections of the fraternity have questioned the motives of the paediatrician, insinuating a desire for visibility or personal gain. This distracts from the real issue. It is far easier to question the intent than engage with the issue raised.

The larger problem is not difficult to recognise. Most professionals remain focused on clinical work and are understandably disengaged from institutional matters. Others, navigating leadership pathways, often operate within networks that favour consensus over confrontation, rewarding accommodation over dissent.

And yet, progress in medicine has rarely relied on unquestioned consensus. It has relied on individuals willing to challenge prevailing practices, often at personal and professional cost. Speaking up in such situations is rarely comfortable, and seldom without consequence.

VITAL CHECK

The issue at hand is not about just one product or company. ORS remains one of the most effective and widely endorsed interventions in public health. Its strength lies in its simplicity and the precision of its formulation. Any dilution or misrepresentation of ORS, whether through marketing or labelling, undermines decades of progress in reducing child mortality.

Individuals who raise concerns act as a necessary check when institutions hesitate and as a reminder that standards in medicine are not negotiable.

Professional bodies will continue to evolve, and leadership will change. But the willingness to question, to challenge and uphold what is right — even when inconvenient — remains the true test of professional integrity. When speaking up becomes the exception rather than the norm, the consequences extend far beyond any single controversy.

The writer is past national convener, IAP Committee on Immunization

CHENNAI
businessline.
MONDAY - MAY 4 - 2026

QUICKLY.

Trump casts doubts on Iran's peace proposal

US President Donald Trump suggested that Iran's latest peace proposal might not be enough to satisfy him. Iran's suggestions include setting a one-month deadline on talks for a deal to reopen the Strait of Hormuz and end both the US naval blockade and the fighting in Iran and Lebanon, *Axios* reported.

If such a pact is reached, another month of discussions would then begin in an attempt to reach an agreement on Iran's nuclear programme. Trump said that he'd been briefed on the "concept of the deal", but later cast doubts on whether it would be acceptable to him. BLOOMBERG

Government's electronics push faces a setback

SNAG PERSISTS. Ministry and electronics industry are still under discussion on Six Sigma, design-led manufacturing and local sourcing

S Ronendra Singh
New Delhi

India's electronics push is facing its first real stress test, with the Centre continuing talks with manufacturers even after the lapse of a 15-day deadline. Ashwini Vaishnaw, Minister of Electronics and Information Technology (MeitY), had set a deadline to firm up plans on design capability, Six Sigma quality standards, talent development and local sourcing under the Electronics Components Manufacturing Scheme (ECMS).

Vaishnaw had struck a rare hard line, telling companies that if they were not serious about upgrading capabilities, they could "walk out of the scheme", and stressing that "the ball is now in the industry's court" and "we have to be in tango". The deadline itself followed a series of nudges from MeitY. In January, the Minister had asked companies to prioritise building dedicated design teams and adopt Six Sigma practices to stay competitive in global value chains.

On March 30, he formalised that push with a 15-day timeline for firms to submit concrete action plans across key parameters, including product design, quality processes, talent pipelines and local sourcing strategies.

The industry, however, was unable to meet the deadline, prompting the Minister's sharp warning that those unwilling to align could opt out of the scheme altogether.

QUALITY FRAMEWORK

More than a month later, consultations remain ongoing, underlining the gap between policy intent and industry preparedness. The government and industry are still working through the targets, sources told *businessline*, even as the Ministry continues parallel discussions with industry bodies to shape the contours of a quality framework and define job roles in electronics manufacturing.

TWO TO TANGO. The industry was unable to meet the deadline for ECMS requirements, prompting a sharp warning that those unwilling to align could opt out of the scheme BLOOMBERG

"The information is being compiled. Also, we are discussing with industry bodies to evolve quality framework and job roles in electronics manufacturing. We should get this going soon," said a senior government official.

> The Ministry also pulled up the India Cellular and Electronics Association and its member firms at a March event for not adhering to the integrated approach laid out by MeitY for the scheme

The Ministry had also pulled up the India Cellular and Electronics Association (ICEA) and its member firms at a March event for not adhering to the integrated approach laid out by MeitY to build a high-quality, self-reliant electronics ecosystem under the ECMS launched in April 2025.

Industry representatives, however, said that process is there albeit slower than envisaged.

A government-constituted committee has been meeting regularly since early April to work out pathways to strengthen research and development (R&D) within firms, both to qualify under the ECMS and to align Indian manufacturers with global benchmarks.

"We have been meeting regularly since early April to discuss all possibilities. Companies do not want anything to be left unchecked, and they also do not want to miss any opportunity under the scheme," said an industry representative requesting anonymity.

The ECMS, which runs for six years from FY26 to FY32, is aimed at building a deeper domestic electronics component ecosystem and reducing import dependence.

It covers a wide range of segments, including sub-assemblies, bare components such as printed circuit boards, connectors and lithium-ion cells, as well as capital goods and other supply chain inputs.

BACK AND FORTH

Even so, the ongoing back-and forth points to a deeper structural challenge: India's shift from a subsidy-led assembly model to one driven by design capability, quality standards and indigenous value addition is proving far more complex than anticipated, forcing a quiet recalibration between the government's policy ambition and the industry's current capacity to deliver.

Data centre boom lifts network solutions firms in FY26

Vallari Sanzgiri
Mumbai

The data centre boom is breathing new life into the network equipment and solutions segment, with service providers reporting stronger order books and a sharp rise in net profits in FY26. Industry leaders say this momentum is likely to sustain into the next fiscal, driven by continued demand for digital infrastructure.

After a muted performance in FY25, HFCL Ltd, Sterlite Technologies Ltd (STL) and Tejas Networks have reported considerable growth in FY26, with all three companies citing data centre and AI infrastructure demand as the driving force.

Earlier, *businessline* had reported that India stands to gain from the geopolitical shifts in West Asia by emerging as a regional hub for cloud infrastructure, amid increasing demand from hyperscalers.

"The demand for the solution providers can continue for the next couple of years as there is huge demand for the data centre business in India. Whether this demand sustains remains to be seen, but at present there is no disruption in this line," said Kranthi Bathini, Director, Equity Strategy at WealthMills Securities, noting the early phase of the data centre business.

BIG LEAP

HFCL, too, estimated such solutions businesses to contribute about ₹400 crore additional revenue in FY27 and about ₹800 crore in FY28. The company reported an all-time high order book of ₹21,200 crore in Q4 FY26 — a 112 per cent increase compared to FY25 — led by the optical fibre cable business.

Its net profit in Q4 FY26 stood at ₹184.45 crore, overcoming its losses in the same quarter last year, while revenue grew to ₹1,824 crore, an annual growth of 128 per cent.

"Data centre inter-connectivity business will increase our revenue significantly this year, and next year we have a very robust performance in front of us," said Mahendra Nahata, Managing Director at HFCL. Despite the huge growth, the management is now hesitant to take up new orders beyond ₹200-500 crore due to capacity constraints and plans to increase its manufacturing capacities multifold.

IMPRESSIVE ORDERS

STL, too, reported an impressive order book that doubled in FY26 to ₹7,687 crore, up 109 per cent year-on-year from ₹3,672 crore in FY25. Like HFCL, STL also turned its loss from a year ago into a net profit of ₹59 crore. Overall revenue grew by 36.9 per cent to ₹1,441 crore.

"With accelerating AI data centre investments and pipeline visibility, we expect the enterprise and data centre segment to scale up to 30 per cent of revenues in the current fiscal," said Ajay Jhanjhari, Chief Financial Officer.

STL noted that AI workloads were exponentially increasing fibre per rack, further spurring demand.

Tejas Networks' order book stood at ₹1,514 crore, an annual growth of 49 per cent, despite a loss of ₹211 crore and reduced revenue of ₹333 crore due to delayed customer projects in the quarter ended March 31, 2026.

The company received strong traction for its 400G/800G optic fibre solutions across telco and carrier-of-carrier networks to serve the surging bandwidth demand for 5G backhaul, enterprise and data centre connectivity, said Arnob Roy, Managing Director and Chief Executive Officer at Tejas Networks.

"The data centre connectivity is going to scale up significantly to provide for the AI cloud interconnect," said Roy.

Jewellery buyers browse more, buy later as gold volatility bites

Aishwarya Kumar
Bengaluru

Jewellery buyers are beginning to look less like impulse shoppers and more like market watchers.

In an era where window shopping has gone digital, consumers are browsing far more than they are buying, tracking gold prices, comparing designs and shortlisting options across platforms before making a purchase. The result: Strong traffic, but slower conversions.

"You cannot walk into a jewellery store 10 times a month and not buy anything. Online works beautifully for browsing, and the behavioural change we're seeing is that women are browsing a lot more than they are buying — 10 to 30 times more," said Gaurav Singh Kushwaha, Founder and CEO of Bluestone

RESEARCH-DRIVEN

The shift is visible across the industry. Suvankar Sen, MD and CEO of Senco Gold and Diamonds, said consumers today are far more research-driven. "Customers are comparing designs, tracking prices and planning purchases more consciously. Intent remains strong, but the journey has become more research-oriented," he noted.

"Consumers are still emotionally attached to gold, but when prices move sharply, they prefer to wait or buy closer to an occasion," Sen added.

Executives at Tanishq said this surge in browsing is not a negative signal. According to CEO Arun Narayan, online platforms are critical for consumer immersion rather than immediate transactions. "Traffic to our website is growing strongly. It reflects high consumer interest, even if the final purchase may happen elsewhere," he said.

LIGHT APPROACH

There is a visible shift towards lighter, lower-ticket jewellery, such as 9K and 14K daily wear, as well as coins and smaller investments.

Even weddings and festival occasions, traditionally strong triggers, are seeing more tactical behaviour.

The challenge is no longer just driving demand, but converting a well-researched browser into a buyer.